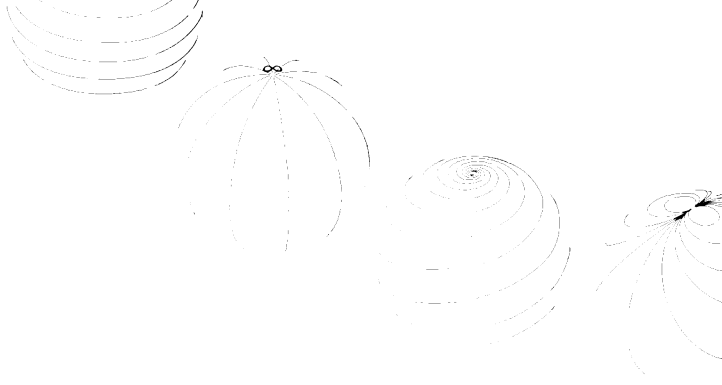




CHAPTER 11

Vector Fields and Complex Integration

11.1 Flux and Work

We promised long ago that there was a more vivid way of understanding complex integrals than the geometric Riemann sum of Chapter 8. In this section we lay the foundations for this elegant new approach. If you are already familiar with vector calculus then you can skip this section and go directly to Section [11.2](#).

11.1.1 Flux

In order to define the flux a little more carefully than before, consider [\[11.1\]](#). At each point of the directed path K we introduce a unit tangent vector $\mathbf{T}$ in the direction of the path, and a unit normal vector $\mathbf{N}$ pointing to our *right* as we travel along K . In terms of the corresponding complex numbers, this convention amounts to

$$\mathbf{T} = i\mathbf{N}.$$

The figure also shows how a vector field $\mathbf{X}$ (which we will first think of as the velocity of a fluid flowing over the plane) can be decomposed into tangential and normal components:

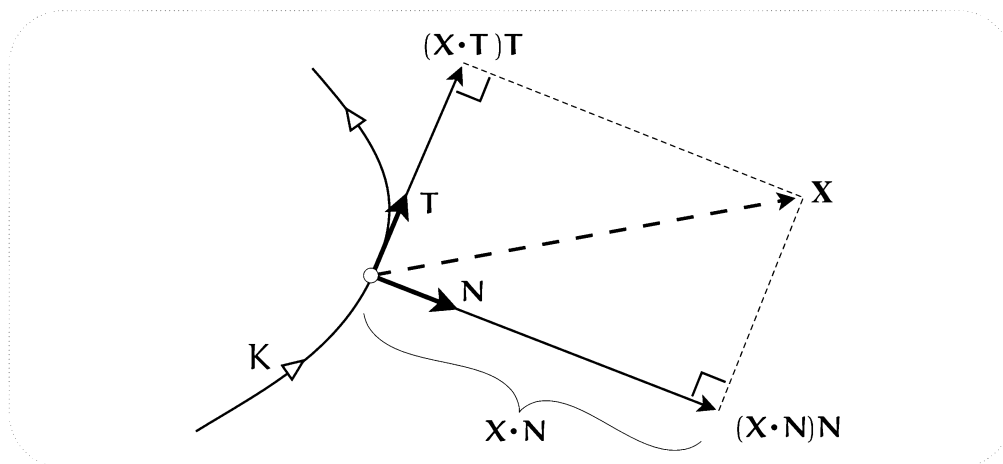
$$\mathbf{X} = (\mathbf{X} \cdot \mathbf{T}) \mathbf{T} + (\mathbf{X} \cdot \mathbf{N}) \mathbf{N}.$$

Only the second of these components carries fluid across K , and the amount flowing across an infinitesimal segment ds of the path in unit time (i.e. its flux) is thus $(\mathbf{X} \cdot \mathbf{N}) ds$. This is a refinement over our previous definition in that the flux now has a *sign*: it is positive or negative according as the flow is from left to right or from right to left. The total flux $\mathfrak{F}[\mathbf{X}, K]$ of $\mathbf{X}$ across K is then the integral of the fluxes across its elements:

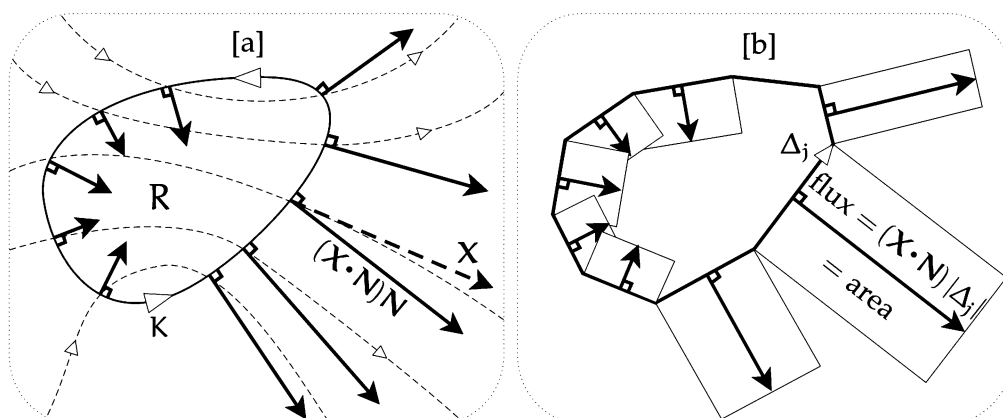
$$\mathfrak{F}[\mathbf{X}, K] = \int_K (\mathbf{X} \cdot \mathbf{N}) ds.$$

Check for yourself that the flux satisfies

$$\mathfrak{F}[-\mathbf{X}, K] = \mathfrak{F}[\mathbf{X}, -K] = -\mathfrak{F}[\mathbf{X}, K].$$



[11.1] Decomposition of a vector into its tangential and normal components. Note that the normal is defined to go to the right of the direction of travel, so that $T = iN$.



[11.2] If we picture X as the velocity of a fluid, then the flux, $\mathfrak{F}[X, K]$, of X across K represents the net flow of fluid out of R per unit time. [a] Mathematically, the flux is the integral of the outward normal component of X . [b] The flux can be visualized as the limit of the sum of the illustrated signed areas: positive if the flow is outward, and negative if it is inward.

The flux concept is further illustrated in [11.2] for the case where K is a simple closed loop bounding the shaded region R . Figure [11.2a] shows the normal components of X , the signed magnitudes of which we must integrate to obtain $\mathfrak{F}[X, K]$. Figure [11.2b] shows how we might make an estimate of this flux. We replace K by a polygonal approximation with directed edges Δ_j , and at the midpoint of each one we draw the normal component of X . The flux is then approximately given by the algebraic sum of the signed areas of the shaded rectangles. In this case there is clearly more positive area than negative, so the flux is positive. As the Δ_j 's become shorter and more numerous, the approximation of course gets better and better.

In the case of the simple loop K in [11.2a] there is another interesting way of looking at the flux:

$$\mathfrak{F}[\mathbf{X}, K] = [\text{fluid leaving } R \text{ per unit time}] - [\text{fluid entering } R \text{ per unit time}].$$

Henceforth we will always take our fluid to be *incompressible*. Thus, provided there are no sources or sinks in R , what flows into R must also flow out of R :

$$\mathfrak{F}[\mathbf{X}, K] = 0.$$

Indeed, we may turn this around and define a flow to be *sourceless* in a region if *all* simple loops in that region have vanishing flux. The simplest example of such a flow without any (finite) sources or sinks is $\mathbf{X} = \text{const.}$ If the loop does contain a source, for example, then incompressibility says that the flux equals the strength of the source.

Although we will only concern ourselves with two-dimensional flows, we should at least mention the concept of flux in three dimensions. If a fluid is flowing through ordinary space, it no longer makes sense to speak of the flux across a curve, but it does make sense to speak of the rate at which the fluid crosses a *surface*. If $\mathbf{N}$ now stands for the normal to this surface, then the flux across an infinitesimal element of *area* dA is once again given by $(\mathbf{X} \cdot \mathbf{N}) dA$. The total flux is then obtained by integrating this quantity over the whole surface. Just as in two dimensions, the incompressibility of a three-dimensional flow is equivalent to the statement that all closed surfaces (that do not contain sources or sinks) have vanishing flux.

Lastly, we should point out that although the word “flux” is Latin for “flow”, it is standard practice to retain this terminology when applying our mathematical definition to *any* vector field $\mathbf{X}$, regardless of whether it actually is the velocity of a flowing substance. For example, the electric field represents a force, but one of the four fundamental laws of electromagnetism says that we can think of it as an incompressible flow in which positive and negative electric charges act as sources and sinks, so that its flux through a closed surface in space equals the net charge enclosed.

11.1.2 Work

So far we have only studied the normal component of $\mathbf{X}$; we turn next to its tangential component. To do so, let us now imagine that $\mathbf{X}$ is a *force field* rather than a flow.

If a particle on which a force acts is displaced infinitesimally then we know from elementary physics that the work done by the field (i.e. the energy it expends) is the component of the force in the direction of displacement, times the distance moved. Thus if the particle moves an amount ds along K then the work done by $\mathbf{X}$ is $(\mathbf{X} \cdot \mathbf{T}) ds$. As with flux, this definition contains a *sign*, the physical significance of

which we will explain shortly. If the particle is moved along the entire length of K , the total work done by the field is then

$$\mathfrak{W}[X, K] = \int_K (X \cdot T) \, ds.$$

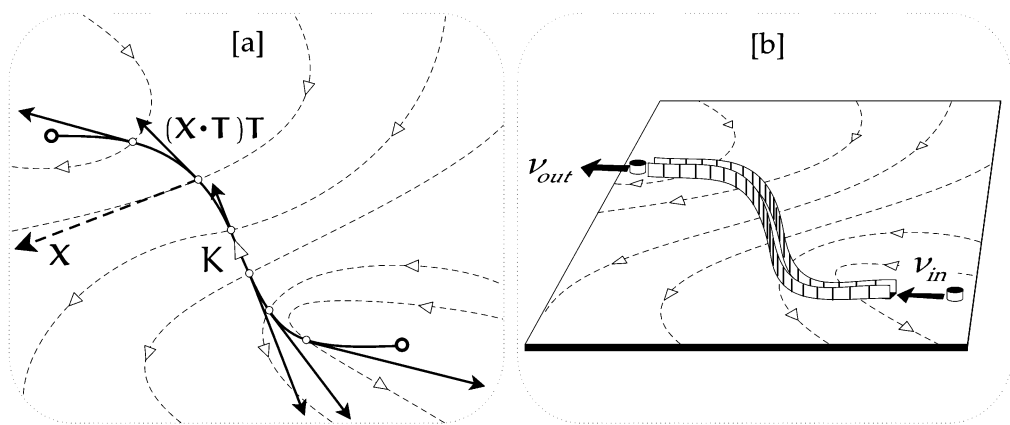
Figure [11.3a] illustrates the tangential components of X on K , the signed magnitudes of which we must add up to obtain $\mathfrak{W}$.

Just as for $\mathfrak{F}$, check that $\mathfrak{W}$ satisfies

$$\mathfrak{W}[-X, K] = \mathfrak{W}[X, -K] = -\mathfrak{W}[X, K].$$

Note that, unlike $\mathfrak{F}$, no modification of $\mathfrak{W}$ is needed if we wish to extend the idea to three-dimensional force fields: it still makes perfectly good sense to consider the work done by the field as a particle is moved along a curve in space, and the formula is as before.

Figure [11.3b] illustrates a simple thought-experiment for interpreting both the magnitude and the sign of $\mathfrak{W}$. We imagine that the plane in which the force field acts is made of ice on which a very small ice-puck of mass m can slide without friction. We now construct a narrow frictionless channel in the shape of K , just wide enough to accommodate the puck which we fire into it with speed v_{in} . On the initial leg of the journey we see that the force opposes the motion, and thus if v_{in} is not sufficiently great, the puck will slow, stop, and return whence it came. Clearly, though, if we fire the puck with sufficient speed it will overcome all resistance and emerge at the end of the channel, say with speed v_{out} . Let the initial and final kinetic energies of the puck be $\mathcal{E}_{in}$ and $\mathcal{E}_{out}$, so that



[11.3] [a] If we picture X as a force field, then the integral of the tangential component is the work, $\mathfrak{W}[X, K]$, done in moving a particle along K . [b] If a puck of mass m is fired with speed v_{in} into a frictionless channel following K , and it emerges with speed v_{out} , then the conservation of energy implies that the work is the change in the kinetic energy: $\mathfrak{W} = \frac{1}{2}mv_{out}^2 - \frac{1}{2}mv_{in}^2$.

$$\mathcal{E}_{\text{in}} = \frac{1}{2}mv_{\text{in}}^2 \quad \text{and} \quad \mathcal{E}_{\text{out}} = \frac{1}{2}mv_{\text{out}}^2.$$

One of the most sacred principles of physics is the “conservation of energy”, which states that energy can never be created or destroyed, only transformed from one kind into another. Thus the energy $\mathfrak{W}$ expended by the force field on the puck does not disappear but instead is transformed into the change in the puck’s kinetic energy:

$$\begin{aligned} \mathfrak{W}[\mathbf{X}, \mathbf{K}] &= \mathcal{E}_{\text{out}} - \mathcal{E}_{\text{in}} \\ &= \left[\frac{m v_{\text{out}} + m v_{\text{in}}}{2} \right] (v_{\text{out}} - v_{\text{in}}) \\ &= [\text{average momentum}] (\text{change in speed}). \end{aligned}$$

This formula also gives clear meaning to the sign of the work: it is the sign of the change in speed. Thus if $\mathfrak{W}$ is positive the field expends energy speeding up the puck and increasing its kinetic energy, while if $\mathfrak{W}$ is negative then the puck has to give up some of its kinetic energy in doing work against the field.

Next, imagine that we bend $\mathbf{K}$ round so that the ends almost join to form a closed loop. When the puck travels along the corresponding channel it will therefore emerge at essentially the same place that it entered. Suppose it were to emerge with greater speed than it entered. Joining the ends of the channel together, the puck would therefore go round the loop faster and faster, gaining energy with each circuit—energy that could be harnessed to solve the world’s energy crisis!

Although we may construct mathematical examples for which this happens, if no energy is supplied from outside the puck/field system then a physical force field will not behave in this way; it will conserve energy so that the puck returns to its starting point with exactly the same speed with which it was launched¹. Such a field is called *conservative*. Mathematically, $\mathbf{X}$ is conservative if and only if

$$\mathfrak{W}[\mathbf{X}, \text{any closed loop}] = 0. \quad (11.1)$$

Just as we applied the concept of flux to vector fields that were not flows, so we may apply the concept of work to vector fields that do not represent force. However, in this general setting it is standard practice to call $\mathfrak{W}[\mathbf{X}, \mathbf{K}]$ the *circulation* of $\mathbf{X}$ along $\mathbf{K}$ rather than the work. As with “flux”, this terminology originates from thinking of $\mathbf{X}$ as representing a flow. To see why, take $\mathbf{K}$ to be a closed loop and consider the following thought-experiment of Feynman (1963). Imagine that the fluid flowing over the plane with velocity $\mathbf{X}$ is instantaneously frozen everywhere

¹ If energy can be supplied from outside the system, then the work need not vanish for a closed loop. In fact the operation of all electrical machines depends on the ability of a moving magnet to create an electric field that *can* speed up our puck. However, there is still no violation of energy-conservation since work is being done to move the magnet. See Feynman (1963).

except within the narrow strip where our channel used to be. The “circulation” is then [exercise] the speed with which the unfrozen fluid flows (or circulates) round K , times the length of K .

If this circulation vanishes for every closed loop then the flow is said to be *irrotational*. Just as “circulation” means $\mathfrak{W}[\mathbf{X}, K]$, irrespective of the physical nature of $\mathbf{X}$, so with equal generality “irrotational” is short for the mathematical statement (11.1). Thus a conservative force field could also be described as irrotational.

11.1.3 Local Flux and Local Work

At present our definition of a *sourceless and irrotational* vector field $\mathbf{X}$ is that

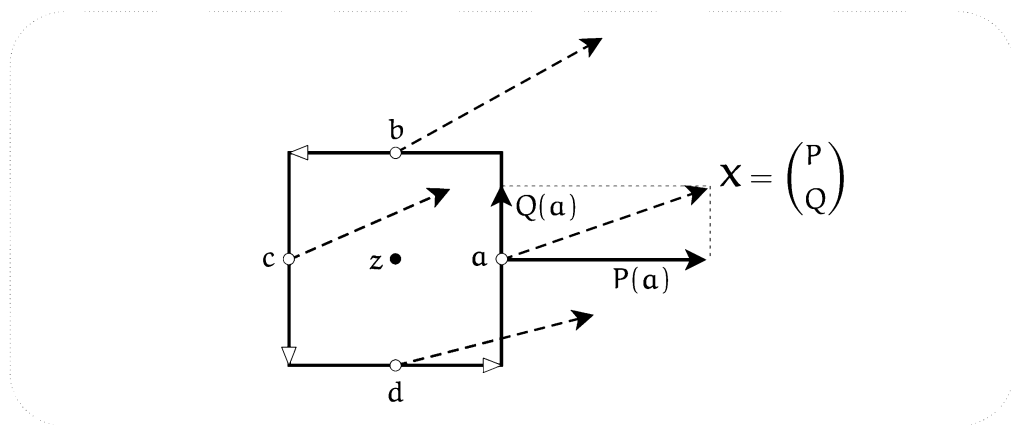
$$\mathfrak{F}[\mathbf{X}, \text{any closed loop}] = 0 \quad \text{and} \quad \mathfrak{W}[\mathbf{X}, \text{any closed loop}] = 0. \quad (11.2)$$

Our next objective is to show that there are two very simple *local* properties of $\mathbf{X}$ that are equivalent to the non-local ones above.

To do this we must calculate limiting behaviour of the flux and the work for a small loop that shrinks to nothing, i.e., for an “infinitesimal loop”. Though it is not entirely obvious, later we will show that this limiting behaviour is independent of the *shape* of the infinitesimal loop. We are thus free to simplify the calculations by choosing the loop to be a small square centred at the point of interest, say z , and having horizontal and vertical edges of length ϵ . See [11.4].

Accurate estimates of $\mathfrak{F}$ and $\mathfrak{W}$ can now be found by evaluating $\mathbf{X}$ at the midpoints (a, b, c, d) of the sides, then summing the appropriate components. In the limit that ϵ shrinks to nothing, this approximation becomes exact, as does the following equation, which we will need in a moment:

$$P(a) - P(c) \asymp \epsilon \partial_x P(z),$$



[11.4] Local flux and local work. Accurate, ultimately exact estimates of $\mathfrak{F}$ and $\mathfrak{W}$ around a small, ultimately vanishing square can be found by evaluating $\mathbf{X}$ at the midpoints (a, b, c, d) of the sides, then summing the appropriate components. Writing $\mathbf{X} = P + iQ$, we find that $\mathfrak{F}[\mathbf{X}, \square] \asymp (\partial_x P + \partial_y Q)(\text{area of } \square)$, and $\mathfrak{W}[\mathbf{X}, \square] \asymp (\partial_x Q - \partial_y P)(\text{area of } \square)$.

where $\partial_x P(z)$ means $\partial_x P$ evaluated at z .

For the flux we find

$$\begin{aligned}\mathfrak{F}[\mathbf{X}, \square] &\asymp \epsilon P(a) + \epsilon Q(b) - \epsilon P(c) - \epsilon Q(d) \\ &\asymp \epsilon \{P(a) - P(c)\} + \{Q(b) - Q(d)\} \\ &\asymp \epsilon^2 [\partial_x P(z) + \partial_y Q(z)].\end{aligned}$$

This expression can be simplified by considering the formal dot product of the *gradient operator* ∇ with the vector field:

$$\nabla \cdot \mathbf{X} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \cdot \begin{pmatrix} P \\ Q \end{pmatrix} = \partial_x P + \partial_y Q.$$

This quantity $\nabla \cdot \mathbf{X}$ is called the *divergence* of $\mathbf{X}$, and in terms of it we have

$$\mathfrak{F}[\mathbf{X}, \square] \asymp [\nabla \cdot \mathbf{X}(z)] (\text{area of } \square). \quad (11.3)$$

In the next section we will see that (11.3) is true if $\square$ is replaced by an infinitesimal loop of arbitrary shape. This important result explains the term “divergence”, for it says that $\nabla \cdot \mathbf{X}$ is the local flux per unit area flowing *away* from z , i.e., diverging from z . In future we will abbreviate “local flux per unit area” to “flux density”.

Repeating the above analysis for the work, we find [exercise]

$$\mathfrak{W}[\mathbf{X}, \square] \asymp [\nabla \times \mathbf{X}(z)] (\text{area of } \square), \quad (11.4)$$

where the formal cross product is defined by

$$\nabla \times \mathbf{X} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \times \begin{pmatrix} P \\ Q \end{pmatrix} = \partial_x Q - \partial_y P.$$

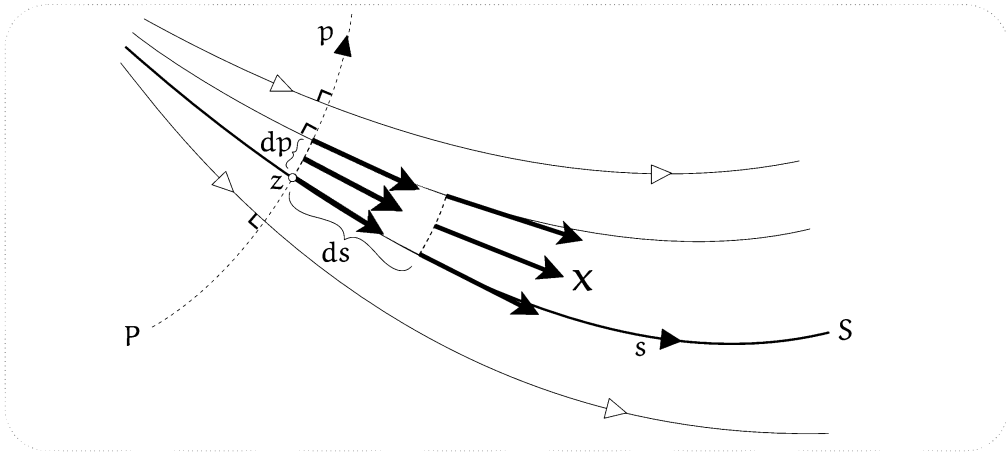
The quantity $\nabla \times \mathbf{X}$ is called the *curl* of $\mathbf{X}$. Geometrically, it measures the extent to which $\mathbf{X}$ ‘curls around’ the point z . Physically, in terms of force fields, the above result says that the curl is the local work per unit area, or work density. There is also a vivid interpretation in terms of flows. If we drop a small disc of paper onto the surface of the flowing liquid at z , in general it will not only start to move (translate) along the streamline through z with speed $|\mathbf{X}(z)|$, but it will also rotate about its centre with some angular speed $\omega(z)$. It can be shown that the aspect of $\mathbf{X}$ which determines the rate of rotation ω is none other than the curl:

$$\omega(z) = \frac{1}{2} [\nabla \times \mathbf{X}(z)].$$

For this reason “curl” is sometimes denoted “rot”, which is short for “rotation”.

11.1.4 Divergence and Curl in Geometric Form*

The above expression for the divergence was obtained by considering the flux out of a shape having no connection with the flow. Greater insight is gained by



[11.5] Divergence and curl in geometric form. Let P be the illustrated orthogonal trajectory of the streamlines, let p measure distance along it, and let s denote distance along the streamline S . Also, let κ_P and κ_S denote the curvatures of P and S , respectively. Then $\nabla \cdot \mathbf{X} = \partial_s |\mathbf{X}| + \kappa_P |\mathbf{X}|$ and $\nabla \times \mathbf{X} = -\partial_p |\mathbf{X}| + \kappa_S |\mathbf{X}|$.

considering the flux out of an infinitesimal “rectangle” R , two sides of which are segments of streamlines of $\mathbf{X}$, while the other two sides are segments of orthogonal trajectories through the streamlines. See [11.5].

Here z is the point down to which R will ultimately be collapsed in order to find the divergence there, S and P are the streamline and orthogonal trajectory through z , and s and p are arc-length along S and P , the direction of increasing p being chosen to make a positive right angle with $\mathbf{X}$.

The net flux out of R is the difference between the fluxes entering and leaving. The flux entering is $|\mathbf{X}| dp$, while the flux leaving is the same expression evaluated on the opposite side of R . It is now clear that two factors contribute to more fluid leaving than entering: (1) greater fluid speed $|\mathbf{X}|$ as the fluid exits; (2) greater separation dp of the streamlines as the fluid exits.

The second factor is clearly governed by how much the direction of $\mathbf{X}$ changes along dp , in other words, by the *curvature* κ_P of P at z . More precisely, if δ denotes the increase in a quantity as we move ds along the streamline, then [exercise] $\delta(dp) = \kappa_P ds dp$. Thus

$$\begin{aligned} (\text{Net flux out of } R) &= \delta\{|\mathbf{X}| dp\} \\ &= (\delta|\mathbf{X}|) dp + |\mathbf{X}| \delta(dp) \\ &= (\partial_s |\mathbf{X}| + \kappa_P |\mathbf{X}|) (\text{area of } R). \end{aligned}$$

The flux density is therefore

$$\nabla \cdot \mathbf{X} = \partial_s |\mathbf{X}| + \kappa_P |\mathbf{X}|. \quad (11.5)$$

In fact [exercise] this formula is still true for a three-dimensional vector field, provided that there exists² a *surface* P orthogonal to the streamlines, and κ_P is taken to be the sum of its principal curvatures.

Turning to the circulation round R , identical reasoning yields [exercise] an equally neat formula for the curl:

$$\nabla \times \mathbf{X} = -\partial_P |\mathbf{X}| + \kappa_S |\mathbf{X}|, \quad (11.6)$$

where κ_S is the curvature at z of the streamline S .

Although we suspect that (11.5) and (11.6) must have been known to the likes of Maxwell, Kelvin, or Stokes, we have not found any reference to these formulae in modern literature.

11.1.5 Divergence-Free and Curl-Free Vector Fields

From the definition (11.2) and the results (11.3) and (11.4) it follows that if $\mathbf{X}$ is sourceless and irrotational throughout some region R , then at each point of R we have

$$\nabla \cdot \mathbf{X} = 0 \quad \text{and} \quad \nabla \times \mathbf{X} = 0.$$

The vector field is then said to be *divergence-free* and *curl-free* in R .

For example, consider the vector field of a point source with strength S :

$$\mathbf{X}(z) = \frac{S}{2\pi z} \iff \mathbf{X} = \frac{S}{2\pi} \begin{pmatrix} x/(x^2 + y^2) \\ y/(x^2 + y^2) \end{pmatrix}.$$

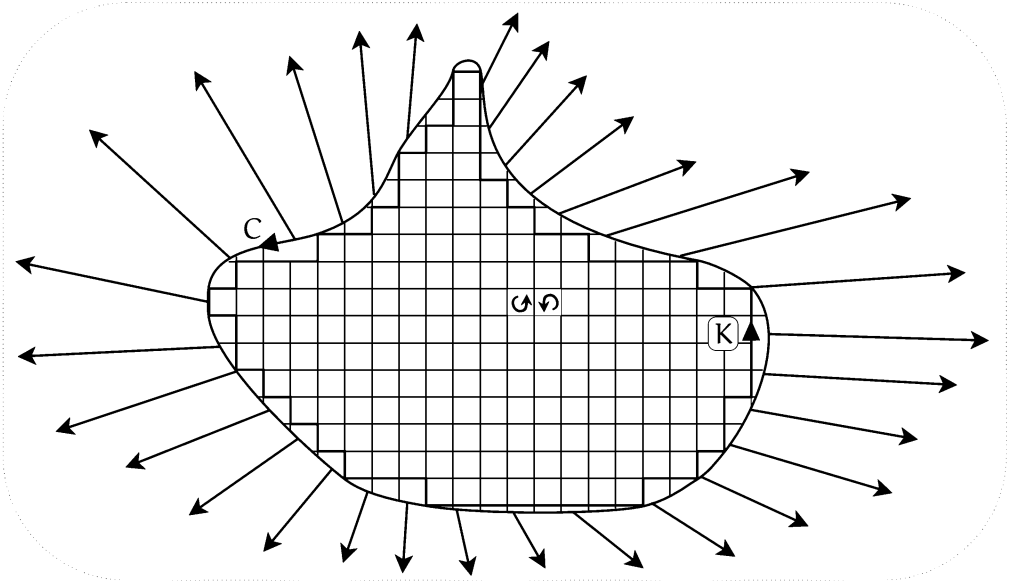
This should have zero flux density (i.e. divergence) everywhere except at the origin, where it should be undefined. Check that this is so. Recall that we previously claimed that this was also the electrostatic field of a long, uniformly charged wire. We can now see that this makes physical sense in that the field is locally conservative. Thus if we fire our puck (which must now carry electric charge in order to experience the force) round an infinitesimal loop, it will return to its starting point with its kinetic energy unchanged. To verify this statement you need only check that the field is curl-free.

We have seen that a sourceless and irrotational field is divergence-free and curl-free. To end this section we wish to establish the converse result: if the divergence and curl vanish throughout a region, the flux and work vanish for all simple loops in that region. We will then have,

A vector field is sourceless and irrotational in a simply connected region if and only if it is divergence-free and curl-free there.

To understand this converse, consider [11.6] which essentially reproduces part of [8.27], p. 468. Let us now recycle the line of reasoning associated with that figure.

² The condition for existence is that the curl either vanish or be orthogonal to the vector field.



[11.6] Gauss's Theorem and Stokes's Theorem. Using the same logic as in the proof of Cauchy's Theorem, summing integrals of the normal or the tangential component of $\mathbf{X}$ around the boundaries of the shaded squares yields the integral along the outer boundary K . But then the flux and work around each square can be expressed in terms of the divergence and curl of $\mathbf{X}$, via (11.3) and (11.4). This immediately yields Gauss's Theorem (11.7) and Stokes's Theorem (11.8).

We begin by noting that as the grid gets finer and finer, the flux or work for K becomes the flux or work for C . Next we relate these quantities to the divergence and curl inside K . Check for yourself that exactly the same mathematical reasoning which previously yielded

$$\oint_K f(z) dz = \sum_{\text{shaded squares}} \oint_{\square} f(z) dz,$$

now yields

$$\mathfrak{F}[\mathbf{X}, K] = \sum_{\text{shaded squares}} \mathfrak{F}[\mathbf{X}, \square]$$

and

$$\mathfrak{W}[\mathbf{X}, K] = \sum_{\text{shaded squares}} \mathfrak{W}[\mathbf{X}, \square].$$

However, in the present context these results become accessible to physical intuition. The first says that the total amount of fluid flowing out of K is the sum of fluxes out of the interior squares. What does the second one say?

Now let the squares of the grid shrink so as to completely fill the interior R of C . Using (11.3) and (11.4) and replacing the sum over squares by a double integral over infinitesimal areas dA , we obtain *Gauss's Theorem*,

$$\mathfrak{F}[\mathbf{X}, C] = \iint_{\mathbf{R}} [\nabla \cdot \mathbf{X}] \, dA, \quad (11.7)$$

and *Stokes's Theorem*,

$$\mathfrak{W}[\mathbf{X}, C] = \iint_{\mathbf{R}} [\nabla \times \mathbf{X}] \, dA. \quad (11.8)$$

From these we see that if the divergence and curl vanish everywhere in $\mathbf{R}$ then the flux and work for C also vanish, as was required.

Again following the logic in Chapter 8, consider what happens to the flux and work as we continuously deform a closed contour, or an open contour with fixed end points. You should be able to see that (11.7) and (11.8) imply two deformation theorems:

If the contour sweeps only through points at which the divergence vanishes, the flux does not change. (11.9)

If the contour sweeps only through points at which the curl vanishes, the work does not change. (11.10)

11.2 Complex Integration in Terms of Vector Fields

11.2.1 The Pólya Vector Field

Consider

$$\int_{\mathbf{K}} H(z) \, dz$$

from the vector field point of view. See [11.7]. In forming a Riemann sum with terms $H \, dz$ we now have the minor advantage that $H = |H| e^{i\beta}$ and $dz = e^{i\alpha} ds$ are not drawn in separate planes, as they were in Chapter 8. However, we still face the problem that $H \, dz = |H| e^{i(\alpha+\beta)} ds$ involves the *addition* of angles, which is not easy to visualize. Just as it is more natural to subtract vectors [yielding connecting vectors] than to add them, so it is also more natural to subtract angles, for this yields the angle contained between two directions.

The simple and elegant solution to our problem is to consider a new vector field: instead of drawing $H(z)$ at z we draw its *conjugate* $\overline{H(z)} = |H| e^{-i\beta}$. We shall call this the *Pólya vector field* of H . Before showing how this solves our problem, let us offer (i) a caution and (ii) a reassurance:

- (i) The Pólya vector field of H is *not* obtained by reflecting the picture of the ordinary vector field for H in the real axis, for this would attach $\overline{H(z)}$ to $\bar{z}$ instead of z . This will become very clear if you (or your computer) draw the Pólya vector fields of z and z^2 , for example. Comparison with [10.1], p. 512, reveals that the resulting phase portraits (not the vector fields themselves) are identical to

those of $(1/z)$ and $(1/z^2)$. This is because $\overline{z^n}$ points in the same direction as $(1/z^n)$.

- (ii) As we will see in a moment, much is gained by representing H by its Pólya vector field, but we also wish to stress that nothing is lost: the new field contains exactly the same information as the old one. For example, it is clear that the index of a loop L merely changes sign when we switch to the Pólya vector field:

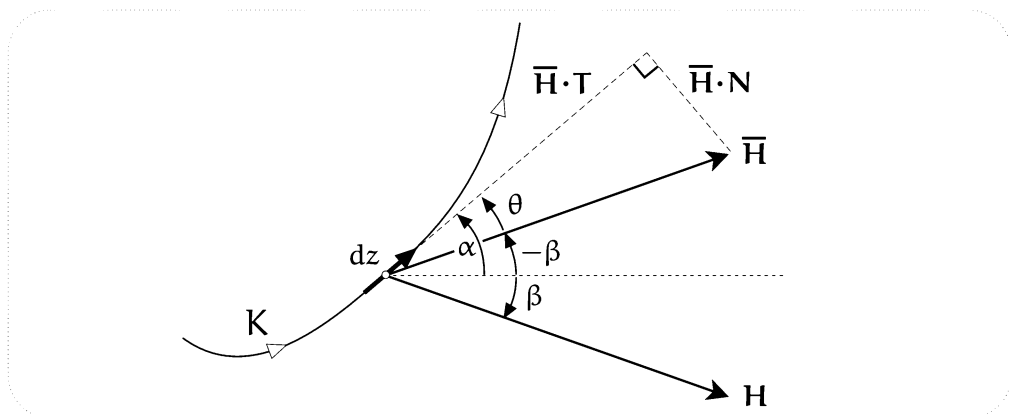
$$\mathcal{I}_{\overline{H}}[L] = -\mathcal{I}_H[L].$$

Thus an n^{th} order root of an analytic H still shows up clearly in its Pólya vector field as a singular point, but now with index $-n$ instead of n . Likewise, a pole of order m produces a singular point of index m instead of $-m$.

Returning to integration, the great advantage of the Pólya vector field is that the angle θ that it makes with the contour (see [11.7]) is given by $\theta = \alpha - (-\beta)$, and this is precisely the angle we were trying to visualize—the angle of the term $H dz$ in the Riemann sum. Better still, we find that

$$\begin{aligned} H dz &= |\overline{H}| e^{i\theta} ds \\ &= [|\overline{H}| \cos \theta + i |\overline{H}| \sin \theta] ds \\ &= [\overline{H} \cdot \mathbf{T} + i \overline{H} \cdot \mathbf{N}] ds. \end{aligned}$$

Thus the real and imaginary parts of each term in the Riemann sum are the work and flux of the Pólya vector field for the corresponding element of the contour. We have thus discovered a vivid interpretation (due to Pólya³ and first championed



[11.7] How the Pólya vector field $\overline{H}$ makes it possible to visualize $\int_K H(z) dz$. We can see that $H dz = [\overline{H} \cdot \mathbf{T} + i \overline{H} \cdot \mathbf{N}] ds$, so this immediately yields the extremely important and useful physical and geometrical interpretation of the complex contour integral: $\int_K H(z) dz = \mathfrak{W}[\overline{H}, K] + i \mathfrak{F}[\overline{H}, K]$.

³ See Pólya and Latta (1974).

by Bart Braden⁴.) of the complex integral of H in terms of the work and flux of its Pólya vector field along the contour:

$$\int_K H(z) dz = \mathfrak{W}[\bar{H}, K] + i \mathfrak{F}[\bar{H}, K]. \quad (11.11)$$

This interpretation is rendered particularly useful by the fact that a computer can instantly draw the Pólya vector field of any function you wish to integrate. You can then quickly get a feel for the value of the integral by looking at how much the field flows along and across the contour. For example, the integral of $(\bar{z}^2 z)$ along the line-segment from $1 - i$ to $1 + i$ is clearly a positive multiple of i . Why? For more on the nitty-gritty of estimating integrals with (11.11), see Braden (1987).

Our interest in (11.11) will be less in this practical aspect, and more in its theoretical import: ideas about flows and force fields can shed light on complex integration, and vice versa. In what follows we shall give examples in both directions.

11.2.2 Cauchy's Theorem

Given a picture of the vector field of a complex mapping $H(z) = u + iv$, how can we tell whether or not H is *analytic*? To my knowledge there is no satisfactory answer to this question as posed. However, there is an answer if we instead look at the Pólya vector field, and it is an answer that exhibits a beautiful connection between physics and complex analysis:

The Pólya vector field of H is divergence-free and curl-free if and only if H is analytic. (11.12)

The verification is a simple calculation:

$$\nabla \cdot \bar{H} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \cdot \begin{pmatrix} u \\ -v \end{pmatrix} = \partial_x u - \partial_y v,$$

and

$$\nabla \times \bar{H} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \times \begin{pmatrix} u \\ -v \end{pmatrix} = -(\partial_x v + \partial_y u).$$

Thus the divergence and curl of $\bar{H}$ will both vanish if and only if the Cauchy–Riemann equations are satisfied. Note for future use that these two equations are really two aspects of a single complex equation,

$$i \partial_x H - \partial_y H = \nabla \times \bar{H} + i \nabla \cdot \bar{H}, \quad (11.13)$$

the vanishing of the LHS being the compact form of the CR equations, (5.1) on page 246.

⁴ See Braden (1985), Braden (1987)—which won the MAA's *Carl B. Allendoerfer Award* in 1988—and Braden (1991).

With this connection established, we now have a second, *physical* explanation of Cauchy's Theorem which is scarcely less intuitive than the geometric one in Chapter 8. For if H is analytic everywhere inside a simple loop K bounding a region R , its Pólya vector field in R will have (as a flow) zero flux density and (as a force field) zero work density. This means that there is no net flux of fluid out of R , and that a puck fired round K returns with its kinetic energy unchanged. From (11.11) we see that the integral of H round K must vanish.

A more mathematical version of this physical explanation was given at the end of the last section in terms of the theorems of Gauss and Stokes. Restating that argument in the present context, for a simple loop K bounding a region R , substitution of (11.7) and (11.8) into (11.11) yields

$$\oint_K H(z) dz = \iint_R [\nabla \times \bar{H}] dA + i \iint_R [\nabla \cdot \bar{H}] dA, \quad (11.14)$$

which vanishes if $\bar{H}$ is curl-free and divergence-free in R .

11.2.3 Example: Area as Flux

As a fun and instructive example, let us reconsider the result

$$\oint_K \bar{z} dz = 2iA \quad (11.15)$$

in the light of the physically intuitive theorems of Gauss and Stokes.

Observe that the Pólya vector field of $H(z) = \bar{z}$ is $\bar{H}(z) = z$, which flows radially outwards from the origin, like a source. However, unlike a source, here the speed of the flow *increases* with distance, making it clear that this flow cannot be divergence-free. Indeed, calculating its flux density, we find that

$$\nabla \cdot \bar{H} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 2.$$

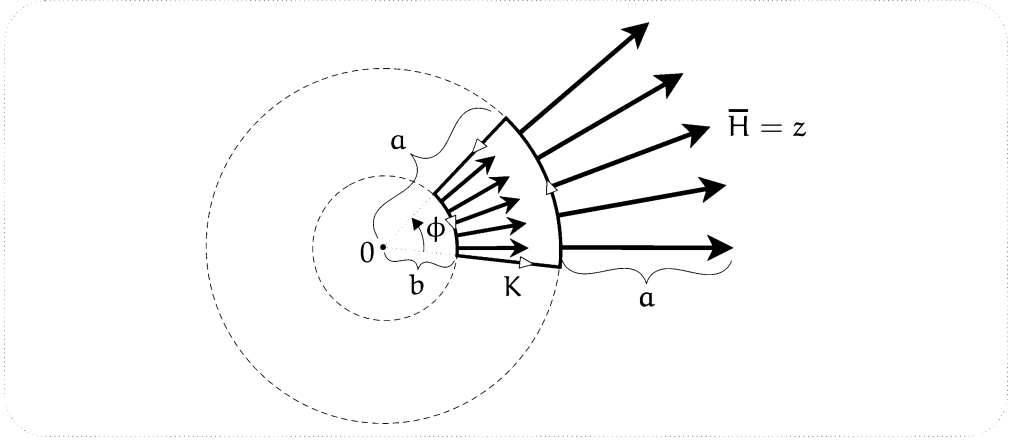
In other words, in each unit of time, 2 units of fluid are being pumped into each unit of area. The flux of fluid out of K is therefore $2A$. On the other hand the flow *is* curl-free:

$$\nabla \times \bar{H} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \times \begin{pmatrix} x \\ y \end{pmatrix} = 0,$$

so there is no circulation round K . Inserting these facts into (11.14) we obtain (11.15).

Figure [11.8] is a concrete example of this new way of looking at (11.15), the shape of K having been chosen so as to make the values of the circulation and flux obvious.

Clearly $\bar{H}(z) = z$ has no circulation along either of the arcs, and it has equal and opposite circulations along the line-segments. The total circulation round K



[11.8] Area as one half the flux of $\overline{H} = z$. Clearly, $\overline{H}(z) = z$ has no circulation along either of the arcs, and it has equal and opposite circulations along the line-segments. The total circulation round K therefore vanishes. Equally clearly, there is no flux across the line-segments, but there is across the arcs. The larger arc has length $a\phi$ and the speed of the fluid crossing it is a , so the flux across it is $a^2\phi$; similarly, for the smaller arc it is $b^2\phi$. Thus, $\oint_K \overline{z} dz = (\text{fluid out}) - (\text{fluid in}) = 2$ (shaded area). Therefore, $\oint_K \overline{z} dz = \mathfrak{W}[z, K] + i \mathfrak{F}[z, K] = 2iA$.

therefore vanishes. Equally clearly, there is no flux across the line-segments, but there is across the arcs. The larger arc has length $a\phi$ and the speed of the fluid crossing it is a , so the flux across it is $a^2\phi$; similarly, for the smaller arc it is $b^2\phi$. Thus,

$$\mathfrak{F}[z, K] = (\text{fluid out}) - (\text{fluid in}) = 2 \left[\frac{1}{2} a^2 \phi - \frac{1}{2} b^2 \phi \right] = 2 \text{ (shaded area)}.$$

Before moving on, let us clear up a paradoxical feature of the vector field z : fluid is being pumped in *uniformly* throughout the plane, and yet the flow appears to radiate from one special place, namely, the origin. The resolution (see [11.9]) lies in the trivial identity $z = z_0 + (z - z_0)$, which says that the flow from the origin is the superposition of the sourceless, irrotational field z_0 and a copy of the original flow, but now centred on the arbitrary point z_0 instead of the origin.

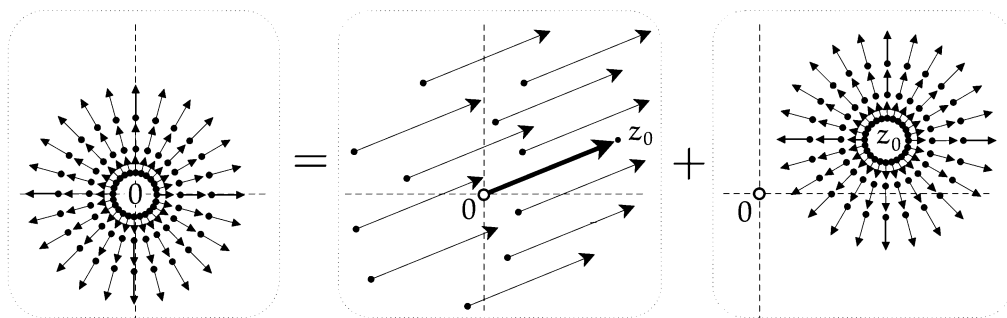
11.2.4 Example: Winding Number as Flux

Next, let us see how the Pólya vector field also breathes fresh meaning into the fundamentally important formula

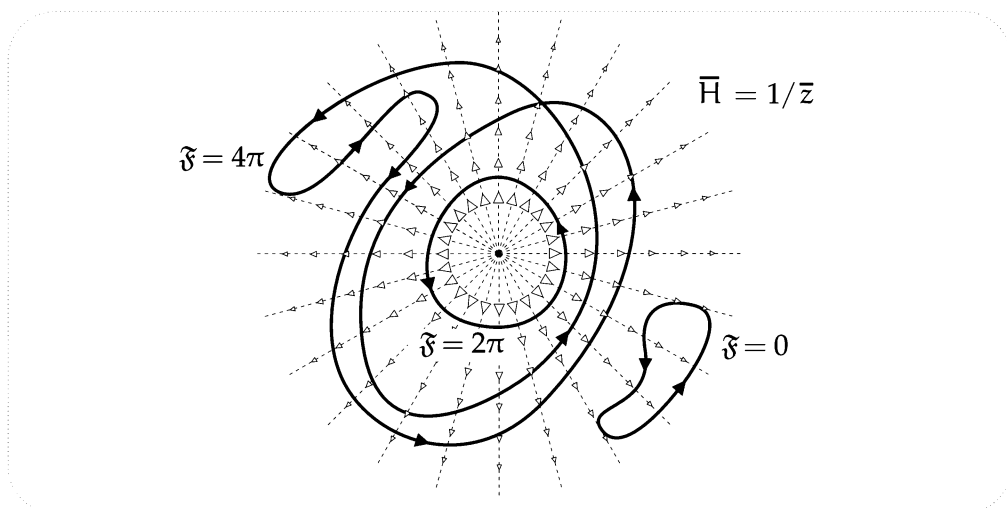
$$\oint_L \frac{1}{z} dz = 2\pi i \nu[L, 0]. \quad (11.16)$$

According to (11.11),

$$\oint_L \frac{1}{z} dz = \mathfrak{W}[(1/\overline{z}), L] + i \mathfrak{F}[(1/\overline{z}), L].$$



[11.9] The identity $z = z_0 + (z - z_0)$ says that the flow from the origin is the superposition of the sourceless, irrotational field z_0 and a copy of the original flow, but now centred on the arbitrary point z_0 instead of the origin.



[11.10] Winding Number interpreted as the flux of a source of strength $(1/2\pi)$. As we discussed in the new Preface, the Pólya vector field of $(1/z)$, namely $(1/\bar{z})$, represents the illustrated source of strength 2π at the origin. Choosing K to be an origin-centred circle K of radius r , traced counterclockwise, we obtain an immediate visual and physical explanation of the iconic fact that $\oint_K \frac{1}{z} dz = 2\pi i$. For clearly there is no flow along K , and since P flows orthogonally across K (from left to right) with speed $(1/r)$, its flux across K is $(2\pi r)(1/r) = 2\pi$. Furthermore, this physical interpretation explains why the value of the integral will remain $2\pi i$ if K is continuously deformed into a general loop encircling the source at the origin, so long as K does not cross that point as it is deformed. And if the loop encircles the source v times, then the intercepted flux is clearly $2\pi v$. Thus if the strength of the source is reduced to $(1/2\pi)$, then v simply is the flux.

But the Pólya vector field $(1/\bar{z})$ is an old friend—it is a source of strength 2π located at the origin.

Figure [11.10] illustrates the intuitive nature of the result from the new point of view. If a loop does not enclose the source, just as much fluid flows out as in; if a

simple loop does enclose the source, it intercepts the full 2π of fluid being pumped in at the origin; more generally, a loop will accrue 2π of flux each time it encircles the source.

To finish the explanation of (11.16) we must show that a source is *pure flux*, i.e. every loop has vanishing work or circulation. Since a source is curl-free except at the origin, Stokes' Theorem guarantees vanishing work for simple loops that do not contain 0. If the loop does contain 0 then it's not so obvious. However, it is obvious for an origin-centred circle. You can now finish the argument for yourself by appealing to the Deformation Theorem (11.10).

In connection with another matter, consider the shaded sector in [11.10]. The same amount of fluid will cross each segment of a contour which passes through it, but the *sign* of the flux will depend on the direction of the contour. Try meditating on the connection between this fact and the crossing rule for winding numbers [(7.1), p. 388].

11.2.5 Local Behaviour of Vector Fields*

We previously showed that $\nabla \cdot \bar{H}$ and $\nabla \times \bar{H}$ represent the flux density and work density of $\bar{H}$ for infinitesimal squares. However, in order for the formulae (11.7) and (11.8) to really make sense it is necessary that these interpretations persist for infinitesimal loops of *arbitrary* shape. Let us now place (11.7) and (11.8) on firmer ground by verifying the shape-independent significance of the divergence and curl. To do so we will first analyse the local behaviour of a general Pólya vector field $\bar{H}$ in the neighbourhood of the origin. The generalization to points other than the origin will be obvious.

As the point $z = x + iy$ moves towards the origin, $H(z)$ will ultimately equal the following formula, in which the partial derivatives are evaluated at 0:

$$\begin{aligned} H(z) - H(0) &\asymp x \partial_x H + y \partial_y H = \frac{1}{2}(z + \bar{z}) \partial_x H - \frac{i}{2}(z - \bar{z}) \partial_y H \\ &= \frac{1}{2} [\partial_x H - i \partial_y H] z + \frac{1}{2} [\partial_x H + i \partial_y H] \bar{z}. \end{aligned}$$

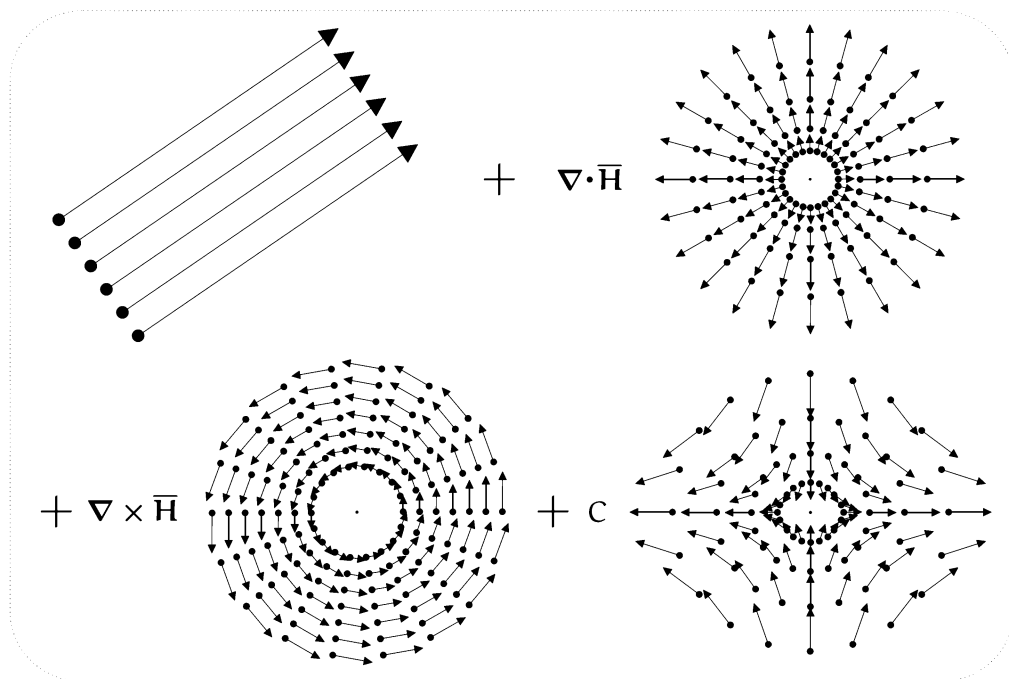
This will become exact in the limit that $|z|$ shrinks to nothing.

Turning to the Pólya vector field itself, and substituting (11.13), we find

$$\overline{H(z)} \asymp \overline{H(0)} + (\nabla \cdot \bar{H}) \frac{z}{2} + (\nabla \times \bar{H}) \frac{iz}{2} + C \bar{z}, \quad (11.17)$$

where $\bar{C} = \frac{1}{2} [\partial_x H - i \partial_y H]$. Note that if H is analytic, in which case $\bar{H}$ is sourceless and irrotational, then (11.17) correctly reduces to the first two terms of Taylor's series: $H(z) = H(0) + H'(0)z + \dots$.

The meaning of the decomposition (11.17) is illustrated in [11.11]. Unless $H(0) = 0$, the constant first term dominates: vectors near the origin differ little from the vector *at* the origin. The remaining three terms correct this crude approximation. The second term describes a vector field (cf. figure [11.8]) that is



[11.11] Local behaviour of a general vector field. In the limit that $z \rightarrow 0$, the general Pólya vector field is given by this ultimate equality: $\bar{H}(z) \asymp \bar{H}(0) + \frac{1}{2}(\nabla \cdot \bar{H})z + \frac{1}{2}(\nabla \times \bar{H})iz + C\bar{z}$. Unless $H(0) = 0$, the constant first term dominates: vectors near the origin differ little from the vector at the origin. The remaining three terms correct this crude approximation. The second term describes a vector field (cf. [11.8]) that is irrotational and has constant divergence, equal to that of $\bar{H}$ at the origin. The third term describes a vector field that is sourceless and has constant curl, equal to that of $\bar{H}$ at the origin. The final term is both irrotational and sourceless.

irrotational and has constant divergence, equal to that of $\bar{H}$ at the origin. The third term describes a vector field that is sourceless and has constant curl, equal to that of $\bar{H}$ at the origin. The final term is both irrotational and sourceless.

Note that this decomposition is geometrically meaningful because the appearance of each of the component vector fields is qualitatively unaffected by the value of its coefficient⁵. We hope these observations make the formula (11.17) both plausible and meaningful.

Now let us return to the original problem. Let K be a small simple loop of arbitrary shape round the origin, and let A be the area it encloses. We wish to show that the divergence and curl of $\bar{H}$ at 0 are the limiting values of the flux per unit area and work per unit area as K shrinks to the origin. Using (11.17) in (11.11) we find

⁵ This is obvious for the source and vortex terms, but not for the last term; see Ex. 10.

$$\begin{aligned}
& \mathfrak{W}[\bar{H}, K] + i \mathfrak{F}[\bar{H}, K] \\
&= \oint_K H(z) dz \\
&= H(0) \oint_K dz + \frac{1}{2} [\nabla \cdot \bar{H} - i \nabla \times \bar{H}] \oint_K \bar{z} dz + \bar{C} \oint_K z dz.
\end{aligned}$$

This becomes exact as K shrinks to nothing. But even if K is not small, we know that the exact values of these three integrals are

$$\oint_K dz = 0, \quad \oint_K \bar{z} dz = 2iA, \quad \oint_K z dz = 0.$$

Thus

$$\mathfrak{W}[\bar{H}, K] + i \mathfrak{F}[\bar{H}, K] = [\nabla \times \bar{H} + i \nabla \cdot \bar{H}] A.$$

Equating real and imaginary parts, we obtain the desired results.

11.2.6 Cauchy's Formula

The Pólya vector field also allows us to cast the mathematical explanation of Cauchy's Formula into a form that is more accessible to physical intuition.

Consider the function

$$H(z) = \frac{f(z)}{(z-p)},$$

where $f(z)$ is analytic. Since H is analytic except at p , its Pólya vector field $\bar{H}$ will have vanishing flux and circulation densities except at p . Thus if C is a simple loop round p , all of its flux and circulation must have originated at p . To find $\mathfrak{W}[\bar{H}, C]$ and $\mathfrak{F}[\bar{H}, C]$ we should therefore examine $\bar{H}$ in the immediate vicinity of p .

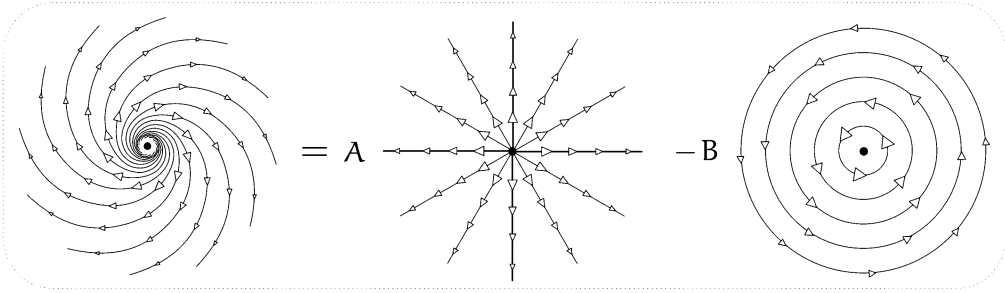
If $f(p) = A + iB$, then very close to p the Pólya vector field $\bar{H}$ will be indistinguishable from

$$\frac{A - iB}{\bar{z} - \bar{p}} = A \left[\frac{1}{\bar{z} - \bar{p}} \right] - B \left[\frac{i}{\bar{z} - \bar{p}} \right].$$

Figure [11.12] illustrates this field for positive A and B , as well as showing the geometric significance of the algebraic decomposition above.

The first term is familiar as a source at p of strength $2\pi A$, a negative value for A corresponding to a sink. The second term is a multiple of the less familiar field $i/(\bar{z} - \bar{p})$ which represents a *vortex*⁶ at p . It is easy to see that the circulation round one of its circular streamlines is 2π , so this will also be its value for any simple loop round p —we say that the vortex has strength 2π . On the other hand its flux vanishes for all loops. While a source is pure flux, a vortex is pure circulation.

⁶ We are now using this term in a narrow sense—previously “vortex” referred to all vector fields of this topological form.



[11.12] A more physical view of Cauchy's Formula. If $f(p) = A + iB$, then as $z \rightarrow p$, the local behaviour of $H(z) \equiv f(z)/(z-p)$ is $H(z) \asymp A[1/(\bar{z}-\bar{p})] - B[i/(\bar{z}-\bar{p})]$, as illustrated. The first term is a source (pure flux) of strength $2\pi A$. The second term represents a vortex (pure circulation) such that the circulation round one of its circular streamlines is $-2\pi B$. If C is a simple loop around p , then $\oint_C \frac{f(z)}{(z-p)} dz = \mathfrak{W}[\bar{H}, C] + i \mathfrak{F}[\bar{H}, C] = -2\pi B + i 2\pi A = 2\pi i f(p)$.

These observations give us a more physical view of Cauchy's Formula:

$$\begin{aligned} \oint_C \frac{f(z)}{(z-p)} dz &= \mathfrak{W}[\bar{H}, C] + i \mathfrak{F}[\bar{H}, C] \\ &= -2\pi B + i 2\pi A \\ &= 2\pi i f(p). \end{aligned}$$

11.2.7 Positive Powers

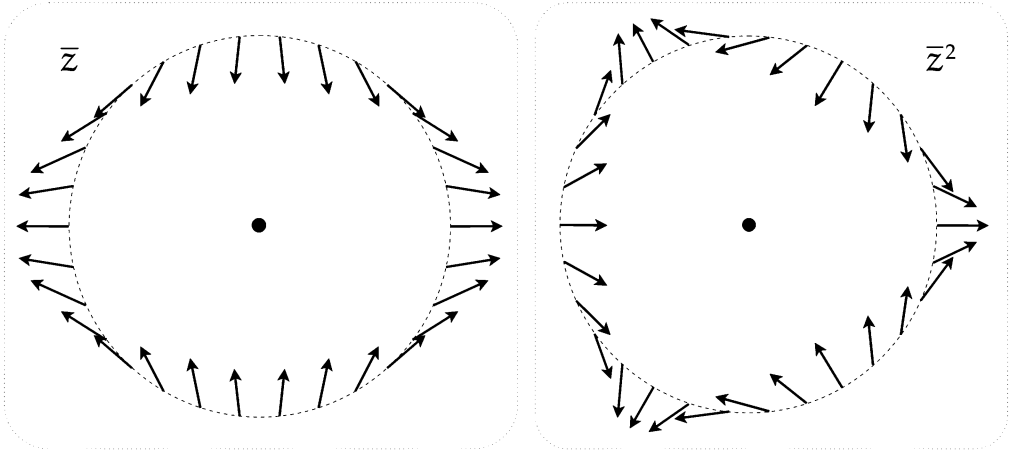
If n is a positive integer then z^n is analytic everywhere and its Pólya vector field $\bar{z}^n$ is correspondingly divergence-free and curl-free. Our physical version of Cauchy's Theorem therefore gives

$$\oint_C z^n dz = \mathfrak{W}[\bar{z}^n, C] + i \mathfrak{F}[\bar{z}^n, C] = 0.$$

At least in the case of an origin-centred circle we can make this much more vivid⁷. Figure [11.13] illustrates the behaviour of $\bar{z}$ and $\bar{z}^2$ on such a circle. It now seems clear that as much fluid flows into each shaded disc as flows out, so that $\mathfrak{F} = 0$, and also (when viewed as force fields) that no net work is done in transporting a particle round the boundary of each disc, so that $\mathfrak{W} = 0$.

We can make this idea precise. First note that for any vector field on the circle, the work and flux will not change their values if we perform an arbitrary rotation of the diagram about the centre of the circle. Next, let us exploit the attractive symmetries of these particular vector fields. Rotating the picture of $\bar{z}$ through $(\pi/2)$ clearly

⁷ In the particular case of z^2 this has also been observed by Braden (1991), though he did not supply the general argument which follows.



[11.13] Pólya vector field explanation of $\oint_C z^n dz = \mathfrak{W}[\bar{z}^n, C] + i \mathfrak{F}[\bar{z}^n, C] = 0$. Draw the Pólya vector field $\bar{z}^n$ of z^n on an origin-centred circle. Rotating this picture about the origin through any angle cannot change either the work or the flux. But rotating the picture by $\pi/(n+1)$ yields the negative of the original field, changing the sign of both the work and the flux. Since the work and the flux must remain unchanged and reverse sign, they must both vanish.

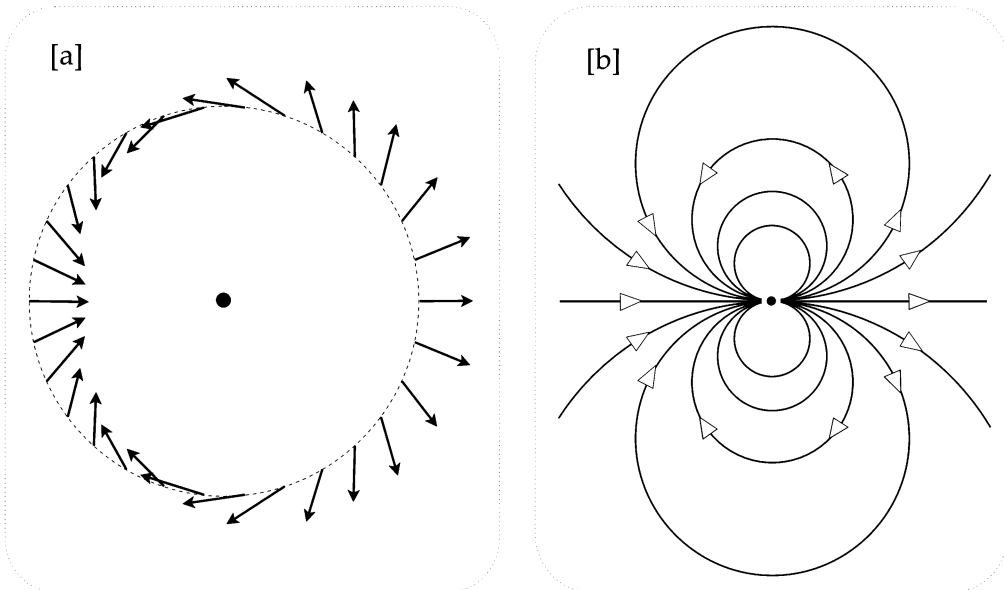
yields the negative of the original field and, correspondingly, the negative of the original work and flux. Since $\mathfrak{W}$ and $\mathfrak{F}$ are simultaneously required to remain the same *and* to reverse sign, they must both vanish.

The same argument applies to $\bar{z}^2$ under a rotation of $(\pi/3)$, and to $\bar{z}^n$ under a rotation of $\pi/(n+1)$. Use your computer to check this for $n = 3$. To understand this symmetry better, consult Ex. 10.

11.2.8 Negative Powers and Multipoles

Consider the negative power functions $(1/z^m)$, where m is a positive integer. Their Pólya vector fields $(1/\bar{z}^m)$ will be divergence-free and curl-free except at the singularity at the origin. Thus if a simple loop C does not enclose the origin, its circulation and flux will vanish. However, since we know from the case $m = 1$ that singularities *are* capable of generating flux and circulation, it remains something of a mystery that (except for $m = 1$) $\mathfrak{W}$ and $\mathfrak{F}$ also vanish if C *does* enclose the singularity.

In the case of an origin-centred circle we can visualize this result exactly as for positive powers. Figure [11.14a] illustrates this for the so-called *dipole* field $(1/\bar{z}^2)$. The argument is also the same as before: this vector field is reversed under a rotation of π , and for $(1/\bar{z}^m)$ it is reversed under a rotation of $\pi/(m-1)$. Knowing that $\mathfrak{W}$ and $\mathfrak{F}$ vanish for the circle tells us [see (11.9), (11.10)] that they will continue to vanish for any loop into which we may deform the circle without crossing the origin.



[11.14] The Dipole Pólya vector field ($1/\bar{z}^2$). [a] If we draw the field on a circle, then a rotation of π reverses the field, so by the same reasoning as the previous figure, $\oint_C (1/z^2) dz = \mathfrak{W}[(1/\bar{z}^2), C] + i \mathfrak{F}[(1/\bar{z}^2), C] = 0$. In general, $(1/\bar{z}^m)$ is reversed under a rotation of $\pi/(m-1)$, so $\oint_C (1/z^m) dz = 0$. [b] The full phase portrait of the dipole field. Note that the streamlines are perfect circles.

Let us now go beyond this geometric explanation in search of a compelling *physical* explanation. Figure [11.14b] shows the phase portrait of the dipole ($1/\bar{z}^2$), the streamlines of which are apparently circular; a simple geometric argument [exercise] confirms their perfect circularity. Where have we seen something like this before? Answer: the doublet field consisting of a source and sink of equal strength S (see [10.3], p. 514). It therefore looks as though we can obtain the dipole simply by coalescing the source and sink. This solves our mystery in a surprising and elegant fashion: neither the source nor the sink generate circulation, and a loop enclosing both receives equal and opposite fluxes.

This explanation is essentially correct. However, as the sink and source move closer and closer together, a greater and greater proportion of the fluid from the source is swallowed up by the sink before it can go anywhere, and at the moment of coalescence the source and sink annihilate each other, leaving no field at all. Let us investigate this algebraically using (10.1), p. 518.

Suppose that the source and sink approach the origin along a fixed line L making an angle ϕ with the real axis. This streamline of symmetry L is called the *axis* of the doublet. Putting $A = \epsilon e^{i\phi} = -B$, the doublet field (10.1) becomes

$$D(z) = \left[\frac{2\epsilon S e^{-i\phi}}{2\pi} \right] \frac{1}{(\bar{z}^2 - \epsilon^2 e^{-i2\phi})}, \quad (11.18)$$

which dies away as the source/sink separation 2ϵ tends to zero. The solution is to increase the strength S in inverse proportion to the separation 2ϵ , so that $2\epsilon S$ remains constant. If we call this real constant $2\pi k$, the limiting doublet field (as $\epsilon \rightarrow 0$) is

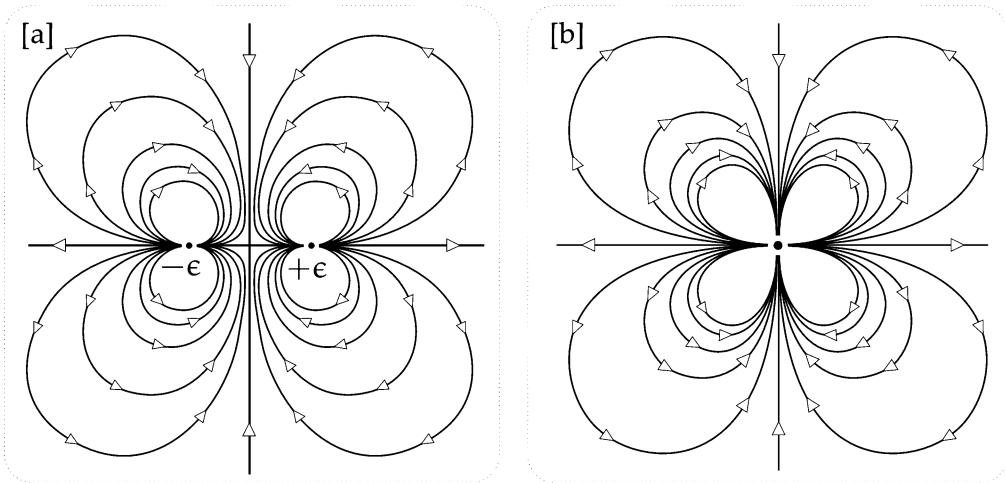
$$D(z) = \frac{k e^{-i\phi}}{\bar{z}^2},$$

i.e. the general dipole field obtained by rotating [11.14] by $+\phi$ and scaling up the speed of the flow by k , which we may think of as the “strength” of the dipole. Thus the Pólya vector field of (d/z^2) is a dipole whose axis points in the direction of d , and whose strength is $|d|$. The complex number d is called the *dipole moment*.

We created the dipole by coalescing equal and opposite sources, increasing their strength so as to avoid mutual annihilation. Continuing this game, we ask, “What will happen if we coalesce equal and opposite *dipoles*, increasing their strength so as to avoid mutual annihilation?” Figure [11.15] reveals the pleasing answer. Figure [11.15a] represents a pair of equal and opposite dipoles located at $\pm\epsilon$ and having real dipole moments $\pm d$, while [11.15b] is the Pólya vector field of $(1/z^3)$. The resemblance is striking, and we can show algebraically that [11.15b], which is called a *quadrupole*, is indeed the appropriate limiting case of [11.15a].

The field for [11.15a] is

$$Q(z) = d \left[\frac{1}{(\bar{z} - \epsilon)^2} - \frac{1}{(\bar{z} + \epsilon)^2} \right] = 4d\epsilon \frac{\bar{z}}{(\bar{z}^2 - \epsilon^2)^2}. \quad (11.19)$$



[11.15] The Quadrupole Pólya vector field $(1/\bar{z}^3)$. [a] Dipoles of strength $\pm d$ at $\pm \epsilon$ are merged by letting $\epsilon \rightarrow 0$. But in order to form a quadrupole, the dipole moments d must grow in inverse proportion to the separation ϵ , so that $d\epsilon$ remains constant, otherwise the two opposite dipoles would simply annihilate each other, leaving no field. [b] The full phase portrait of the resulting quadrupole.

Once again letting the strength d grow in inverse proportion to the separation, so that $k = 4d\epsilon$ remains constant, the coalescence of the dipoles yields the quadrupole:

$$Q(z) = \frac{k}{\bar{z}^3}.$$

In general, the Pólya vector field of (q/z^3) is called a quadrupole with *quadrupole moment* q .

We have thus explained the vanishing circulation and flux of $(1/\bar{z}^3)$: each of the dipoles in [11.15a] is known not to generate any circulation or flux, so the quadrupole in [11.15b] won't either. You are invited to continue this line of thought by showing (geometrically and algebraically) that the fusion of two quadrupoles yields the so-called *octupole* field, $(1/\bar{z}^4)$, and so on.

Dipoles, quadrupoles, octupoles, etc., are collectively known as *multipoles*. Similarly, dipole moments, quadrupole moments, etc., are collectively known as *multipole moments*.

11.2.9 Multipoles at Infinity

Although there is no mystery surrounding the vanishing circulation and flux for positive powers, it would still be nice to find a physical explanation analogous to the one for negative powers. To see how this can be done, we begin by considering the constant function $f(z) = a$, the Pólya vector field of which is a flow of constant speed $|a|$ in the direction $\bar{a}$.

Standing in the midst of this flow, the fluid seems to originate far over the horizon in the direction $-\bar{a}$ and to disappear over the horizon in the opposite direction, as though both a source and a sink were present at infinity. To make sense of this idea, stereographically project the streamlines onto the Riemann sphere. Since the streamlines are parallel lines in the direction $\bar{a}$, their projections are circles which all pass through the north pole in the same direction. We thus obtain a picture similar to [10.11b], p. 525: a *dipole at infinity*!

Let us analyse this further. If we were standing at the midpoint of the source and sink of the doublet in [10.3], p. 514, the flow in our vicinity would have approximately constant speed and direction. As the source and sink recede from us towards infinity, ultimately coalescing there to form a dipole, the approximation to a constant field gets better and better. The snag is that in this process the magnitude of the field at any finite point dies away to nothing.

We see this algebraically in (11.18): $D(z) \rightarrow 0$ as $\epsilon \rightarrow \infty$. However, if we let S grow in proportion to the separation, so that $(S/\epsilon) = \text{const.} = k\pi$, say, then as $\epsilon \rightarrow \infty$ the doublet field tends to the constant field $D(z) = -k e^{i\phi}$.

Given that z^0 yields a dipole at infinity, what might the Pólya vector field of z^1 correspond to? Use your computer to see that it is a quadrupole at infinity. Verify this algebraically using (11.19). Continuing in this fashion, one finds [exercise] that z^2 corresponds to an octupole at infinity, and so on.

11.2.10 Laurent's Series as a Multipole Expansion

The above ideas shed new light on the Laurent series and the Residue Theorem. Suppose that an otherwise analytic function $f(z)$ has a triple pole at the origin. We know from Chapter 9 that $f(z)$ will have a Laurent series of the form

$$f(z) = \underbrace{\frac{q}{z^3} + \frac{d}{z^2} + \frac{\rho}{z}}_{P(z)} + a + bz + cz^2 + \dots \quad (11.20)$$

In the vicinity of the singularity, the behaviour of f is governed by its principal part P , the Pólya vector field of which is

$$\overline{P(z)} = \frac{\overline{q}}{\overline{z}^3} + \frac{\overline{d}}{\overline{z}^2} + \frac{\overline{\rho}}{\overline{z}}.$$

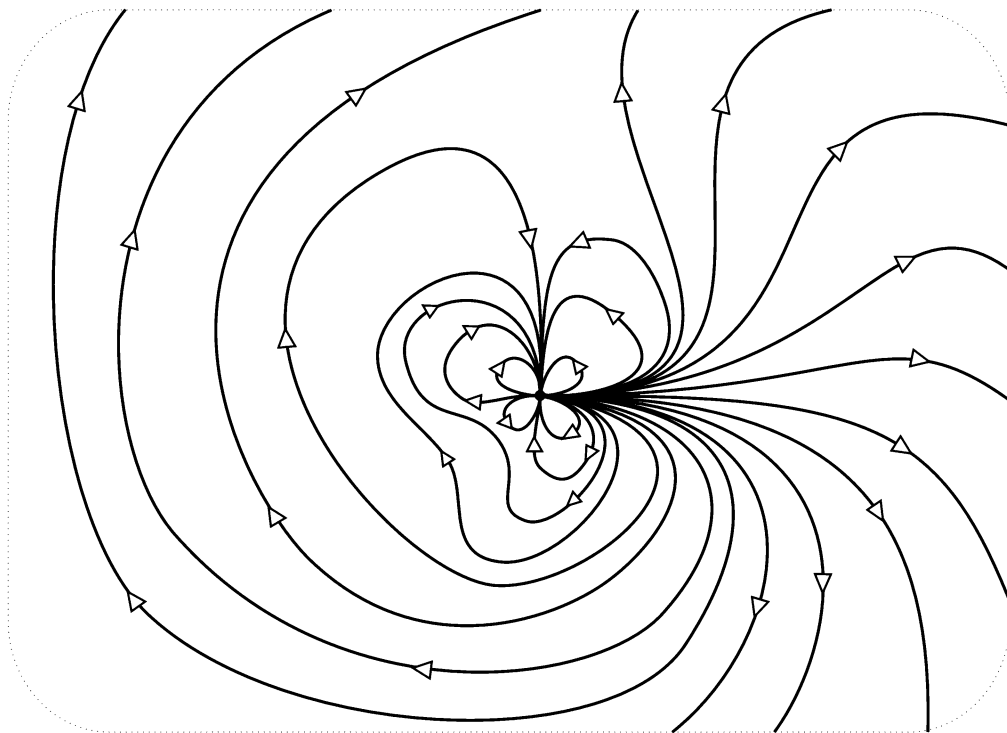
This we now recognize to be the superposition of a quadrupole, a dipole, and a source-vortex combination of the type shown in [11.12]. Thus the principal part of the Laurent series amounts to what a physicist would call a *multipole expansion*.

To visually grasp the meaning of such an expansion, consider [11.16] which illustrates a typical $\overline{P}$. Very close to the singularity, the field is completely dominated by the quadrupole with its characteristic four loops, but as we move slightly further away the quadrupole's influence wanes relative to the dipole. Indeed, at this intermediate range we clearly see the characteristic two loops of a dipole. Finally, at still greater distances, both the quadrupole and the dipole become insignificant relative to the source-vortex, the precise form of which is determined solely by the residue ρ . Compare with [11.12], in which $\rho = A + iB$.

Continuing our outward journey, now well beyond the unit circle, the entire principal part becomes negligible relative to the remaining terms of (11.20). First a becomes important, then bz takes over, and so on. Thus as we approach infinity the field at first resembles a dipole, then a quadrupole, and so on. However, unlike the approach to the pole, on the journey to infinity we may experience multipoles of greater and greater order, without end.

Of course in general f may possess other singularities and (11.20) will cease to be meaningful when $|z|$ increases to the distance of the nearest one. Nevertheless, in the region where it is valid, we may still think of the non-negative powers as representing multipoles at infinity.

To recap, Laurent's series and the Residue Theorem may be conceived of physically as follows. The only term capable of generating circulation and flux is (ρ/z) , which may itself be decomposed into a vortex of strength $\mathfrak{V} = -2\pi \text{Im}(\rho)$ and a source of strength $\mathfrak{F} = 2\pi \text{Re}(\rho)$. All the other terms correspond to multipoles which generate neither circulation nor flux; a finite collection of these reside at the pole, while the rest are at infinity.



[11.16] Multipole expansions and the Residue Theorem. Consider the Pólya vector field $\overline{f(z)}$ of an analytic function in the vicinity of a triple pole. The dominant behaviour near the singularity is given by the conjugate of the principal part of the Laurent series: $\overline{P(z)} = \frac{\overline{q}}{z^3} + \frac{\overline{a}}{z^2} + \frac{\overline{p}}{z}$, which we recognize to be the superposition of a quadrupole, a dipole, and a source–vortex combination of the type shown in [11.12]. Thus the principal part of the Laurent series amounts to what a physicist would call a multipole expansion. Very close to the singularity, the field is completely dominated by the quadrupole with its characteristic four loops, but as we move slightly further away the quadrupole’s influence wanes and we see the characteristic two loops of a dipole. Finally, at still greater distances, both the quadrupole and the dipole become insignificant relative to the source–vortex, the precise form of which is determined solely by the residue ρ . Neither the quadrupole nor the dipole generates any circulation or flux, so if L encircles the pole, the circulation and flux is entirely due to the source–vortex: $\oint_L f(z) dz = 2\pi i\rho$.

11.3 The Complex Potential

11.3.1 Introduction

Phase portraits are so convenient that it is easy to forget that in general they cannot represent the *lengths* of the vectors. In this section we shall see that if a vector field is either sourceless or irrotational (or both) then there exists a special way of drawing the phase portrait so that the lengths *are* represented.

Although we shall ultimately be concerned with the Pólya vector fields of analytic functions, which are both sourceless *and* irrotational, it is more instructive to analyse the implications for sourcelessness and irrotationality separately. Nevertheless, in view of the final objective, we shall continue to write the vector field as $\overline{H}$.

11.3.2 The Stream Function

First let $\overline{H}$ be a sourceless flow of fluid. The Deformation Theorem (11.9) tells us that the flux across a curve connecting two given points is independent of the choice of the curve. Thus if K is any contour from an arbitrary fixed point a to a variable point z , the flux across it, namely

$$\Psi(z) \equiv \mathfrak{F}[\overline{H}, K],$$

will be a well-defined function of z , called the *stream function*. If we choose a different point a then the new stream function will only differ from the old one by an additive constant.

Suppose that z lies anywhere on the streamline through a . See [11.17]. Choosing K to be the portion of the streamline from a to z , we see that $\Psi(z) = 0$. Similarly, suppose that q lies anywhere on a streamline through another point p . Taking K to be a path from a to p , followed by the section of the streamline from p to q , we see that $\Psi(q) = \Psi(p)$. In other words,

The streamlines are the level curves of the stream function Ψ .

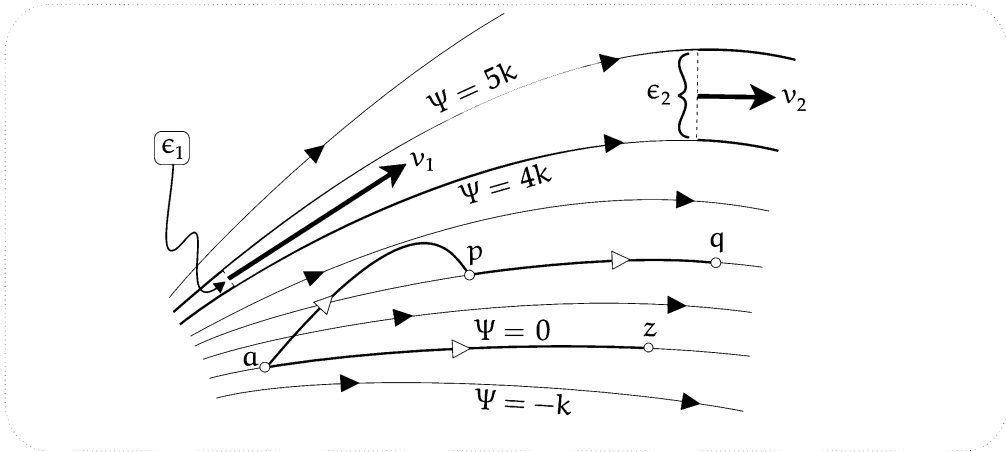
Instead of constructing the phase portrait by drawing random streamlines, suppose we do it as follows: *choose a number k and draw just those streamlines for which $\Psi = 0, \pm k, \pm 2k, \pm 3k, \dots$* See [11.17]. Having drawn the phase portrait in this special way, the speed of the flow is represented by the crowding together of the streamlines. Let's justify this claim and make it more precise.

Since no fluid crosses the streamlines, we may think of the region lying between two adjacent ones as a tube down which fluid flows. Any curve connecting the two sides will have the same flux, namely k . Adapting the language of Faraday and Maxwell, we may thus describe the tube more quantitatively as a *k -flux tube*.

The shaded area in [11.17] is part of one such tube, the initial and final cross-sections (lengths ϵ_1 and ϵ_2) having been drawn perpendicular to the flow. If k is chosen small, the speed $v = |\overline{H}|$ of the flow will be approximately constant across these ends, say v_1 and v_2 . Thus the fluxes into and out of the shaded region (which must both equal k) are approximately $\epsilon_1 v_1$ and $\epsilon_2 v_2$. As k is chosen smaller and smaller, these expressions become more and more accurate:

$$v_1 = \frac{k}{\epsilon_1} \quad \text{and} \quad v_2 = \frac{k}{\epsilon_2}. \quad (11.21)$$

In order to maintain a constant flux k , the speed must decrease as the tube widens.



[11.17] The Stream Function Ψ and the k -flux tubes of a sourceless flow. If $\bar{H}$ is sourceless, and K is any contour connecting an arbitrary fixed point a to a variable point z , the flux across it is independent of K , yielding the well-defined stream function, $\Psi(z) \equiv \mathfrak{F}[\bar{H}, K]$. Now choose a number k and draw just those streamlines for which $\Psi = 0, \pm k, \pm 2k, \pm 3k, \dots$. Any cross section of a channel between neighbouring streamlines therefore intercepts the same flux k , and so we call it a k -flux tube. This special phase portrait (pioneered by Faraday and Maxwell) has the great advantage that the speed of the flow is now visible via $|\bar{H}| \propto (k/\epsilon)$: the denser the streamlines, the faster the flow.

To summarize:

Let the phase portrait of a sourceless vector field be constructed using k -flux tubes. If k is chosen small, the speed of the flow at any point will be approximately given by k divided by the width of the tubes in the vicinity of the point. For infinitesimal k , the result is exact. (11.22)

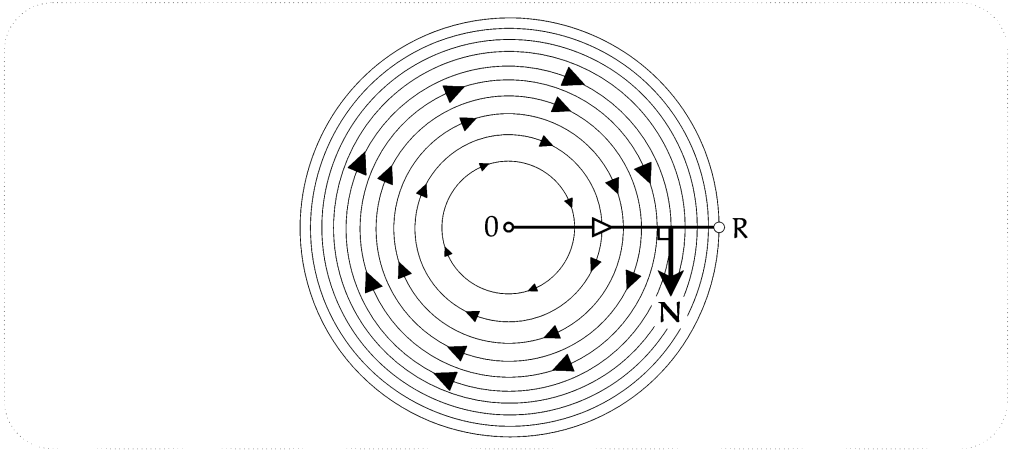
However, since the number of k -flux tubes passing through a given region will vary inversely with k , our phase portrait will get very cluttered if k is chosen too small. In practice (cf. [10.3], p. 514) we get a good feel for the speed of the flow with relatively few streamlines.

Let's apply these ideas to the simple (non-analytic) example $H(z) = i\bar{z}$. The Pólya vector field is then

$$\bar{H} = -iz \iff \bar{H} = \begin{pmatrix} y \\ -x \end{pmatrix},$$

the streamlines of which are clockwise circles round the origin, the speed of the flow round each one being equal to its radius. See [11.18].

Although this vector field is not irrotational [$\nabla \times \bar{H} = -2$], it is sourceless [$\nabla \cdot \bar{H} = 0$], and thus it possesses a stream function. For convenience's sake, let's choose $a = 0$. We already know that the streamlines are origin-centred circles, so to find the value of Ψ on the streamline of radius R we must find the flux for any



[11.18] The Pólya vector field $\bar{H} = -iz$, drawn using k -flux tubes.

path from the origin to any point on this circle. Choosing the path to be the portion of the positive real axis from 0 to R , we see that

$$ds = dx \quad \text{and} \quad \mathbf{N} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

Thus

$$\Psi = \int (\bar{H} \cdot \mathbf{N}) ds = \int_0^R x dx = \frac{1}{2} R^2.$$

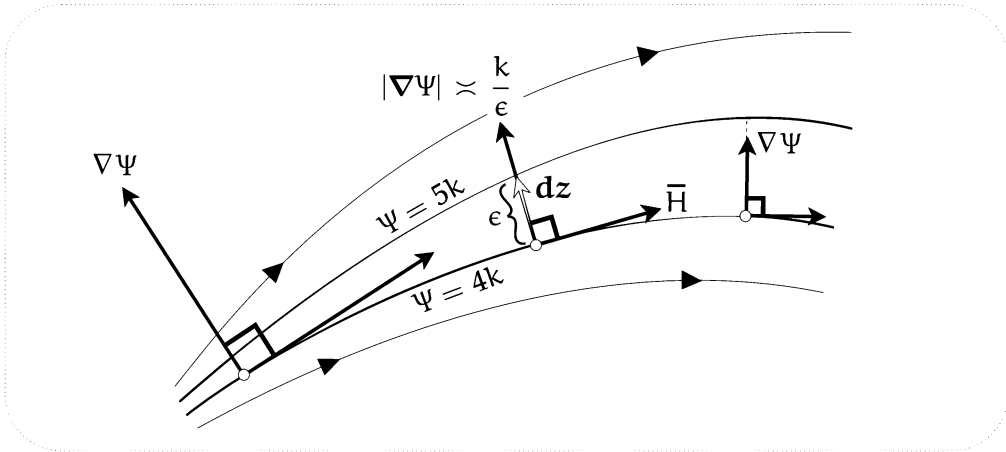
Knowing the stream function we are now in a position to draw the special phase portrait. Choosing $k = (1/2)$ we find that the radii of the streamlines are $\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots$. Figure [11.18] illustrates these streamlines, and qualitatively confirms the prediction of (11.22). As we move outward from the origin the streamlines become more crowded together, reflecting the increasing speed of the flow.

11.3.3 The Gradient Field

We have seen in geometrical terms how it is possible to reconstruct a sourceless vector field $\bar{H}$ from a knowledge of its stream function Ψ . In order to find a simple formula for $\bar{H}$ in terms of Ψ , we need the concept of the *gradient field* $\nabla\Psi$. This is defined to be the vector field

$$\nabla\Psi = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \Psi = \begin{pmatrix} \partial_x \Psi \\ \partial_y \Psi \end{pmatrix} \iff \nabla\Psi = \partial_x \Psi + i \partial_y \Psi.$$

The gradient field $\nabla\Psi$ has a simple geometric interpretation in terms of the streamlines of [11.17]. To see this, we express the infinitesimal change $d\Psi$ resulting from an infinitesimal movement $dz = dx + i dy$ as a dot product:



[11.19] The field in terms of the stream function: $\bar{H} = -i \nabla \Psi$. First, the direction of $\nabla \Psi$ is the one that is orthogonal to the streamlines and along which Ψ increases. Thus $-i \nabla \Psi$ points in the direction of $\bar{H}$. Second, $|\nabla \Psi| \asymp (k/\epsilon) \asymp |\bar{H}|$. Done.

$$d\Psi = (\partial_x \Psi) dx + (\partial_y \Psi) dy = \begin{pmatrix} \partial_x \Psi \\ \partial_y \Psi \end{pmatrix} \cdot \begin{pmatrix} dx \\ dy \end{pmatrix} = \nabla \Psi \cdot dz.$$

If dz is tangent to a streamline then $d\Psi = 0$, so $\nabla \Psi$ has vanishing dot product with this direction. Also, Ψ increases when dz makes an acute angle with $\nabla \Psi$. Thus

The direction of $\nabla \Psi$ is the one that is orthogonal to the streamlines and along which Ψ increases. Thus $-i \nabla \Psi$ points in the direction of $\bar{H}$. (11.23)

See [11.19].

So much for the direction of $\nabla \Psi$; what about its magnitude? In [11.19] (which is basically a copy of [11.17]) we imagine that k is infinitesimal. Choosing $dz = e^{i\theta} ds$ in the direction of $\nabla \Psi$, we find $d\Psi = |\nabla \Psi| ds$. In particular, if we let ds equal ϵ (the width of the k -flux tube) then $d\Psi$ will equal k . Thus

$$|\nabla \Psi| \asymp (k/\epsilon).$$

But this is precisely the formula we previously obtained for the speed $v = |\bar{H}|$ of the flow! Thus $|\nabla \Psi| = |\bar{H}|$.

Combining this result with (11.23) we obtain the following simple formula for $\bar{H}$ in terms of Ψ :

$$\bar{H} = -i \nabla \Psi \iff \bar{H} = \begin{pmatrix} \partial_y \Psi \\ -\partial_x \Psi \end{pmatrix}. \quad (11.24)$$

Try this out on our previous example $H(z) = i\bar{z}$, the Pólya vector field of which had stream function $\Psi = (x^2 + y^2)/2$.

Now consider the question, “What additional condition must be satisfied by Ψ if $\bar{H}$ is also required to be *irrotational*?” The answer is that it must satisfy *Laplace’s equation*:

$$\Delta\Psi \equiv \partial_x^2 \Psi + \partial_y^2 \Psi = 0.$$

Solutions of this equation are called *harmonic*, so we may restate the result as follows:

A sourceless field is irrotational if and only if its stream function is harmonic.

The verification is a simple calculation:

$$\nabla \times \bar{\mathbf{H}} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \times \begin{pmatrix} \partial_y \Psi \\ -\partial_x \Psi \end{pmatrix} = -\Delta\Psi.$$

11.3.4 The Potential Function

Next suppose that $\bar{\mathbf{H}}$ is a force field which is known to be conservative (irrotational). In this case it is the work rather than the flux which must be path-independent. Thus if K is any contour from an arbitrary fixed point a to a variable point z , the work done by the field in moving the particle along K is a well-defined function of z ,

$$\Phi(z) = \mathfrak{W}[\bar{\mathbf{H}}, K].$$

This is called the *potential function*, though there are several pseudonyms depending on the context: e.g., in electrostatics it is called the “electrostatic potential”, in hydrodynamics it is called the “velocity potential”, and in the case of flowing heat it is already familiar as the temperature. As with the stream function, changing the choice of a merely changes Φ by an additive constant.

Let’s investigate Φ as we did Ψ . The level curves $\Phi = \text{const.}$ are called *equipotentials*; what is their geometric significance? As [11.20] illustrates, the answer is that

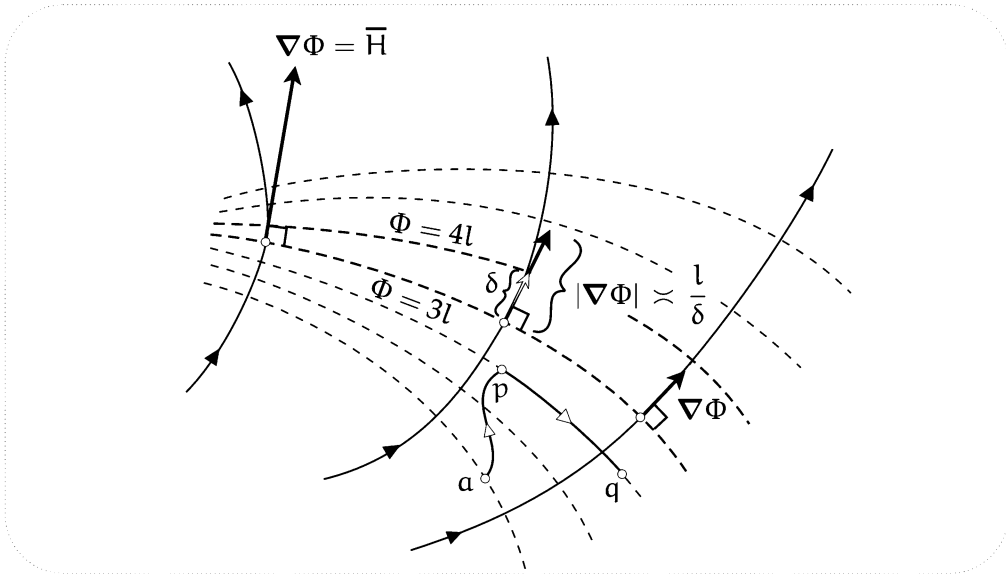
The equipotentials are the orthogonal trajectories through the lines of force. (11.25)

The reason should be clear. A certain amount of work $\Phi(p)$ is done in moving the particle from a to p , but then no additional energy is expended in moving it to q along the orthogonal trajectory through p . Thus $\Phi(q) = \Phi(p)$.

Instead of illustrating random equipotentials, [11.20] mimics the special construction used in [11.17]: *we draw just those equipotentials for which $\Phi = 0, \pm l, \pm 2l, \pm 3l, \dots$* In this picture the same amount of work l is required to move the particle from each equipotential to the next. Let us therefore call the region lying between two such adjacent equipotentials an *l-work tube*.

Suppose that l is chosen small, and consider the work done in moving a particle along the correspondingly short cross-section δ in [11.20]. In the limit of vanishing l we find that

$$|\bar{\mathbf{H}}| = \frac{l}{\delta}. \quad (11.26)$$



[11.20] The Potential Function Φ of a conservative force field. If K is any contour connecting an arbitrary fixed point a to a variable point z , the work done by the conservative (irrotational) field in moving the particle along K is a well-defined function of z , independent of K , called the potential function, $\Phi(z) = \mathfrak{W}[\vec{H}, K]$. Just as we did with the stream function, we draw just those equipotentials for which $\Phi = 0, \pm l, \pm 2l, \pm 3l, \dots$. In this picture the same amount of work l is required to move the particle from each equipotential to the next, so we call the region lying between two such adjacent equipotentials an l -work tube. The equipotentials are the orthogonal trajectories through the lines of force. Also, $|\nabla\Phi| \approx (l/\delta) \approx |\vec{H}|$, so $\vec{H} = \nabla\Phi$.

Thus the magnitude of the force is represented by the crowding together of the equipotentials:

Let the equipotentials of a conservative force field be constructed using l -work tubes. If l is chosen small, the magnitude of the force at any point will be approximately given by l divided by the width of the tubes in the vicinity of the point. For infinitesimal l , the result is exact. (11.27)

Since the gradient field $\nabla\Phi$ is automatically orthogonal to the equipotentials and has magnitude (l/δ) , we may combine (11.25) and (11.27) into the simple formula

$$\vec{H} = \nabla\Phi \iff \vec{H} = \begin{pmatrix} \partial_x \Phi \\ \partial_y \Phi \end{pmatrix}. \quad (11.28)$$

Lastly, suppose that $\vec{H}$ is required to be sourceless. Since

$$\nabla \cdot \vec{H} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \cdot \begin{pmatrix} \partial_x \Phi \\ \partial_y \Phi \end{pmatrix} = \Delta\Phi, \quad (11.29)$$

we see that

A conservative force field is sourceless if and only if its potential function is harmonic. (11.30)

11.3.5 The Complex Potential

We now know two things about a vector field $\bar{H}$ that is irrotational *and* sourceless: (i) both Φ and Ψ exist; (ii) it is the Pólya vector field of an analytic function. In this section we shall attempt to illuminate the connections between these two facts.

Since Φ and Ψ both exist, we may superimpose pictures of types [11.17] and [11.20], thereby simultaneously dividing the flow into mutually orthogonal k -flux tubes and l -work tubes. Before drawing this picture let us *choose the increment of work to be numerically equal to the increment of flux*: $l = k$.

Let us call the intersection of a k -flux tube with a k -work tube a k -cell. We already know that the sides of each k -cell meet at right angles, so for small k they will be approximately rectangles. The sides of such a rectangle will be the previously considered widths ϵ and δ of the two kinds of tube. But combining the results (11.21) and (11.26) we see that

$$\frac{k}{\epsilon} = |\bar{H}| = \frac{k}{\delta} \implies \delta = \epsilon.$$

Thus

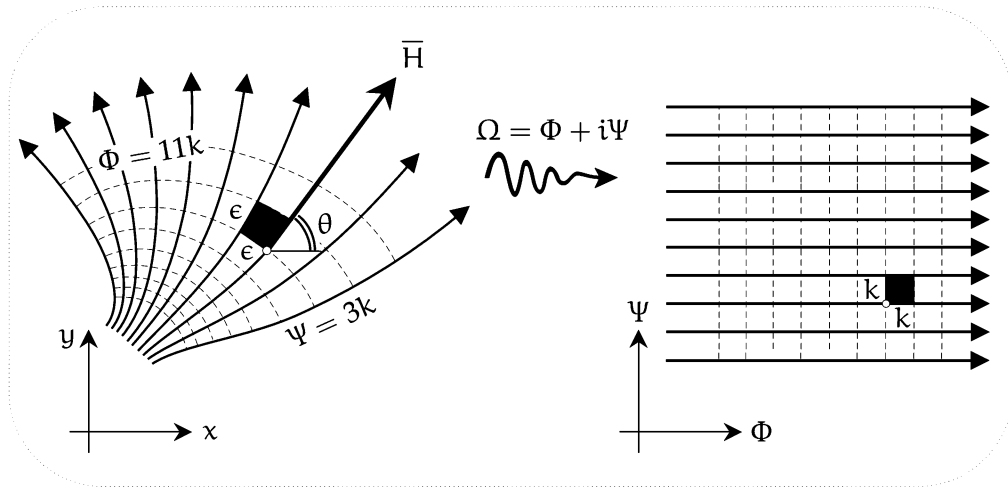
In the limit of vanishing k , the k -cells are squares. (11.31)

The LHS of [11.21] illustrates such a division into approximately square k -cells. We have labelled $\Phi = 11k$ and $\Psi = 3k$, but we have left it to you [exercise] to label the remaining streamlines and equipotentials; this can only be done in one way.

Note that once such a special phase portrait (including the equipotentials) has been drawn with a small value of k , the value of $\int_L H dz$ is easy to find. For if L crosses m equipotentials and n streamlines, an accurate estimate of the integral will be $k(m + in)$. If L crosses an equipotential or streamline more than once, how should m and n be counted?

We mention in passing that there is an interesting physical interpretation of the k -cells which is due to Maxwell (1881). Suppose that the vector field represents the flow of a fluid having unit mass per unit area. In the limit of vanishing k , the speed v will be constant throughout any particular cell, and the kinetic energy of the fluid in that cell will be

$$\text{kinetic energy} = \frac{1}{2}(\text{area}) v^2 = \frac{1}{2}\epsilon^2 \left(\frac{k}{\epsilon}\right)^2 = \frac{1}{2}k^2.$$



[11.21] The Complex Potential, $\Omega = \Phi + i\Psi$. If $H(z)$ is analytic, then $\overline{H(z)}$ is both irrotational and sourceless, so we may simultaneously define and superimpose l -work tubes orthogonal to the streamlines with k -flux tubes along the streamlines, thereby dividing the region of the flow into small “rectangles” (as l and k tend to zero). We now go one step further and choose $l = k$, dividing the region into squares bounded by equipotentials and streamlines. This makes it vividly clear that if we combine the potential and stream functions into the single complex potential $\Omega = \Phi + i\Psi$, then the mapping Ω is locally an amplitwist, i.e., Ω' exists. Indeed, we will show that $H = \Omega'$.

Thus

Each k -cell contains the same amount of energy, and the total energy in a region is thus obtained by counting the number of k -cells contained within it.

If we reinterpret the vector field as an electrostatic field, and correspondingly reinterpret “energy” as electrostatic energy, the result is still valid; this was the context in which Maxwell discovered it.

The result (11.31) is intimately connected with ideas of complex analysis. To see this, let us combine the potential and stream functions into a single complex function Ω called the *complex potential*:

$$\Omega(z) = \Phi(z) + i\Psi(z).$$

Returning to the dominant point of view of this book, think of Ω as a *mapping*. The RHS of [11.21] shows the image of the special phase portrait under this mapping:

The complex potential maps streamlines to horizontal lines and equipotentials to vertical lines. Furthermore, each square k -cell is mapped to a square of side k . Thus Ω is an analytic mapping.

We may check this symbolically. Equating (11.24) and (11.28) we obtain

$$\begin{pmatrix} \partial_x \Phi \\ \partial_y \Phi \end{pmatrix} = \begin{pmatrix} \partial_y \Psi \\ -\partial_x \Psi \end{pmatrix},$$

which are the CR equations for Ω .

What is the amplitwist of the complex potential? By considering the effect of Ω on the black k-cell in [11.21] we see that if the streamline through z makes an angle θ with the horizontal, the twist of Ω at z is $-\theta$, which we recognize as the angle of $H(z)$. We also see that the amplification of Ω is (k/ϵ) , which we recognize as $|\bar{H}| = |H|$. Thus

$$\Omega' = H.$$

Since H is the derivative of an analytic function, it must itself be analytic. We have thus obtained a second, more geometrical proof that the class of sourceless, irrotational vector fields is the same as the class of Pólya vector fields of analytic functions.

The result $\Omega' = H$ can be checked symbolically. Substituting one of the CR equations for Ω into (11.28), we obtain

$$\bar{H} = \nabla\Phi = \partial_x\Phi + i\partial_y\Phi = \partial_x\Phi - i\partial_x\Psi = \overline{\partial_x\Omega} = \overline{\Omega'}.$$

When we thought of an analytic function f as a conformal mapping, f' represented its amplitwist. But since any such function may instead be thought of as the complex potential of a flow, we now have another interpretation of differentiation: f' is the conjugate of the velocity of the flow described by f . Correspondingly, we also have a new interpretation of critical points: they are the places where the velocity vanishes. Such places are called *stagnation points* in the flow.

By analysing the implications of sourcelessness and irrotationality separately, we have been able to understand the Pólya vector fields of non-analytic functions that may possess a stream function or a potential function, but not both. If we had instead restricted ourselves from the outset to the Pólya vector fields of analytic functions, the complex potential could have been obtained more rapidly (but less revealingly) as follows.

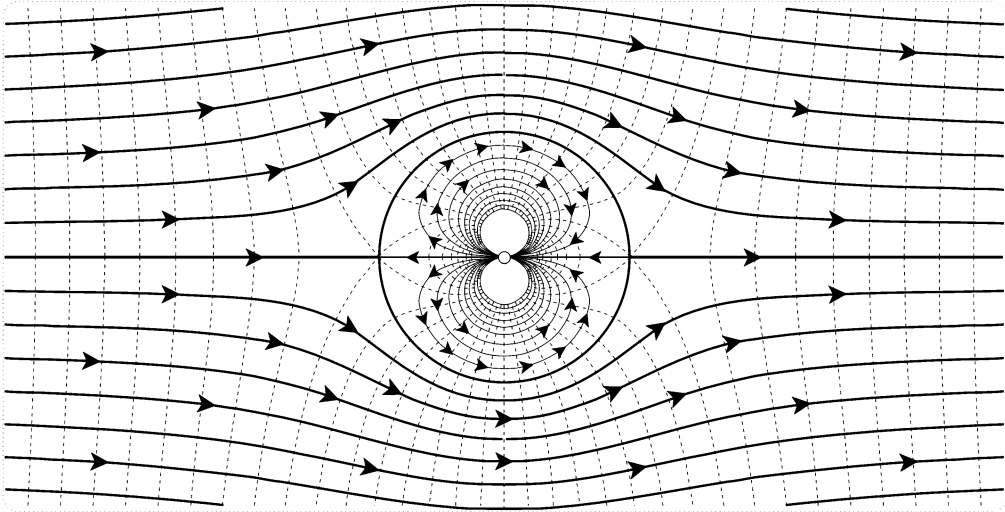
If L is any contour from an arbitrary fixed point a to a variable point z , we may define

$$\Omega_L(z) = \int_L H(w) dw = \mathfrak{W}[\bar{H}, L] + i\mathfrak{F}[\bar{H}, L].$$

But, as we saw in Chapter 8, if H is analytic then this integral is independent of L , and the well-defined function

$$\Omega(z) = \int_a^z H(w) dw = \Phi(z) + i\Psi(z)$$

is in fact the antiderivative of H . More explicitly, the image $\Omega(L)$ of a contour L from p to q is the path taken by the Riemann sum for the integral of H along L . The value of the integral is then the vector connecting the start of $\Omega(L)$ to its finish, namely, $\Omega(q) - \Omega(p)$.



[11.22] $\Omega(z) = z + \frac{1}{z}$ represents the superposition of a uniform eastward flow and a dipole.

11.3.6 Examples

(1) We previously claimed that the streamlines of the dipole $\bar{H} = (1/\bar{z}^2)$ in [11.14b] were perfect circles, and we asked you to provide a simple geometric proof. A second demonstration is obtained by finding the complex potential:

$$\bar{H} = \frac{1}{\bar{z}^2} \implies \Omega' = \frac{1}{z^2} \implies \Omega = -\frac{1}{z} + c.$$

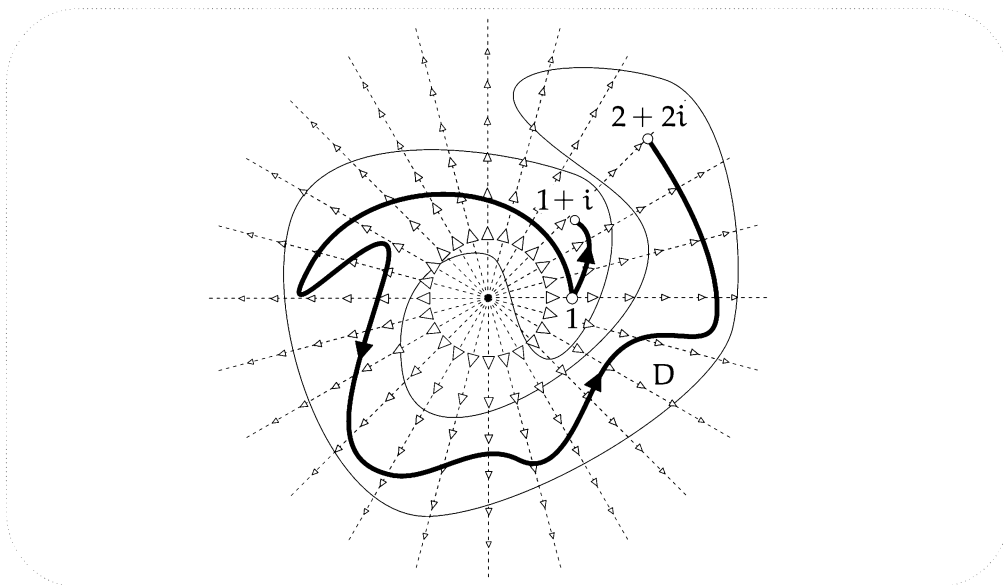
The streamlines are the images under $\Omega^{-1}(z) = -1/(z - c)$ of horizontal lines. The result follows from the fact that inversion sends straight lines to circles through the origin.

(2) A uniform eastward flow has complex potential $\Omega = z$. If we insert a dipole of complex potential $\Omega = (1/z)$ into this flow then the new flow will be the superposition of the two individual flows and thus will have complex potential

$$\Omega(z) = z + \frac{1}{z}.$$

Using your computer you may verify that the streamlines and equipotentials are as shown in [11.22]. Note how the streamlines emanating from the dipole are deformed out of perfect circularity by the uniform flow, but that this distortion diminishes as the origin is approached.

(3) A source of strength 2π at the origin has vector field $\bar{H} = (1/\bar{z})$. If we choose to measure work and flux along a path L emanating from $z = 1$ then [see p. 465] the complex potential is



[11.23] The complex potential of the source $(1/\bar{z})$ is a multivalued function: although the work is single-valued, the flux increases by 2π with every extra revolution around the source. But if we restrict attention to a simply connected region D that does not contain the source, then Ω becomes single-valued.

$$\Omega(z) = \Phi(z) + i\Psi(z) = \log_L(z) = \ln|z| + i\theta_L(z).$$

While the work Φ is single-valued, the flux Ψ is a multivalued function whose values differ from each other by multiples of 2π . This makes perfect sense since each time L encircles the source it intercepts the full 2π of fluid being pumped in there. Note that the single-valued inverse function $\Omega^{-1}(z) = e^z$ does indeed map horizontal and vertical lines to the source's streamlines and equipotentials.

If we wish to obtain a single-valued complex potential we may do so by confining our attention to any simply connected region not containing the source. The shaded region D in [11.23] is an example. Any two paths from 1 to z that lie wholly within D may be deformed into each other without ever leaving D , hence without crossing the source, hence without altering the flux. For example, we see that for the particular choice of D in [11.23], the unique values of Ψ at $(1+i)$ and at $(2+2i)$ are $(\pi/4)$ and $(9\pi/4)$. However, a different choice of D might well yield different values of Ψ at these two points.

More generally, if D is any simply connected region not containing any singularities of an otherwise analytic H , the Pólya vector field $\bar{H}$ will possess a single-valued complex potential in D .

11.4 Exercises

- 1 For each of the following vector fields X verify that the geometric formulae (11.5) and (11.6) yield the correct values for the divergence and for the curl:
 - (i) $X = (1/\bar{z})$.
 - (ii) $X = \bar{z}$.
 - (iii) $X = z^2$, where $z = x + iy$.
 - (iv) $X = y^2$, where $z = x + iy$.
 - (v) $X = i(1/r^2)e^{i\theta}$, where $z = re^{i\theta}$.
- 2 For each of the following vector fields X , calculate $\mathfrak{F}[X, C]$ and $\mathfrak{W}[X, C]$ for the given loop C , then check your answers by substituting the results of the previous question into (11.7) and (11.8).
 - (i) $X = x^2$, and C is the edge of the rectangle $a \leq x \leq b$, $-1 \leq y \leq 1$, traversed counterclockwise.
 - (ii) $X = i(1/r^2)e^{i\theta}$, and C is the edge of the region $a \leq r \leq b$, $0 \leq \theta \leq \pi$, traversed counterclockwise.
- 3 Use a computer to draw the Pólya vector field of $f(z) = 1/[z \sin z]$ and thereby identify the locations and orders of the poles of $f(z)$. For each of the following choices of C , numerically estimate $\oint_C f(z) dz$ by making on-screen measurements of the vectors, then estimating the flux and circulation round C . In each case check your estimate by calculating the exact answer using residue theory.
 - (i) Let C be a small circle centred at $-\pi$.
 - (ii) Let C be a small circle centred at 0 .
 - (iii) Let C be a small circle centred at π .
 - (iv) Let C be a small circle centred at 2π .
 - (v) Let C be the boundary of the rectangle $1 \leq x \leq 7$, $-1 \leq y \leq 1$.
- 4 Repeat parts (i) and (ii) of the previous question using $f(z) = z \operatorname{cosec}^2 z$.
- 5 Let L be a contour from the real number $-\theta$ to $+\theta$. By choosing L to be a line-segment, and then sketching the Pólya vector field at points along L , show that $\int_L z e^{iz} dz$ is purely imaginary. Verify this by calculating the exact value of the integral.
- 6 All complex analysis texts recognize the great utility of the inequality

$$\left| \int_L f(z) dz \right| \leq \int_L |f(z)| \cdot |dz|, \quad (11.32)$$

but none that we know of have sought to answer the question, “When does equality hold?” This is probably because no elegant answer is forthcoming (cf. our attempt in Chapter 8) without the concept of the Pólya vector field. However, armed with the Pólya vector field, we have what we shall call *Braden’s Theorem*⁸:

Equality holds in (11.32) if and only if the contour L cuts the streamlines of the Pólya vector field of f at a constant angle.

Explain Braden’s Theorem.

- 7 Continuing from the previous question, suppose that $f(z) = \bar{z}$.
 - (i) Show that if L is a segment of the spiral with polar equation $r = e^\theta$, then the condition of Braden’s Theorem is met.
 - (ii) Verify by explicit calculation that equality does indeed hold in (11.32), as predicted.

- 8 Consider the flow created by $(2n + 1)$ sources, each of strength 2π , located at

$$0, \pm\pi, \pm2\pi, \dots, \pm n\pi.$$

- (i) If $\Omega_n(z)$ denotes the complex potential of this flow, show that

$$\Omega_n(z) = \ln \left[z \left(1 - \frac{z^2}{\pi^2} \right) \left(1 - \frac{z^2}{2^2\pi^2} \right) \cdots \left(1 - \frac{z^2}{n^2\pi^2} \right) \right] + \text{const.}$$

- (ii) Ignoring the constant, and referring to Ex. 13 on p. 508, deduce that as the number of sources increases without limit, $\Omega_n(z)$ tends to $\Omega(z) = \ln[\sin z]$.
 - (iii) Check that this answer makes sense by using a computer to draw the velocity vector field, $V = \overline{\Omega'}$.
- 9
 - (i) Explain why the derivative of the complex potential of a source yields the complex potential of a dipole.
 - (ii) Referring to the previous question, draw a sketch predicting the appearance of the flow whose complex potential is $\Omega(z) = \frac{d}{dz} \ln[\sin z]$. Check your answer by getting the computer to draw this flow.
- 10 Reconsider the term $C\bar{z}$ in the local decomposition (11.17) of a general vector field. See [11.11].
 - (i) Show that the visual appearance of the vector field $C\bar{z}$ is essentially independent of the value of C . More precisely, show that if $C = e^{i\phi}$ then increasing ϕ merely causes the entire picture of the vector field $C\bar{z}$ to rotate, in fact exactly half as fast as $e^{i\phi}$ rotates.

⁸ See Braden (1987). We independently recognized this fact, probably at about the same time as Braden himself.

- (ii) To make the result vivid, create a computer animation of the vector field $e^{i\phi} \bar{z}$ as ϕ increases from 0 to π .
 - (iii) More generally, show that if n is an integer and $F(z)$ stands for either $\bar{z}^n$ or z^{-n} , then the vector field of $e^{i\phi} F$ is obtained by rotating the vector field of F through $\phi/(n+1)$. [Note that the $n = -1$ fields (including sources and vortices) are exceptional.]
- 11 Consider a flow such that the inverse complex potential is $\Omega^{-1}(w) = w + e^w$. Use a computer to draw the streamlines, and verify mathematically that the picture may be interpreted as the flow out of a channel $-\pi \leq \text{Im}(z) \leq \pi$, $\text{Re}(z) \leq -1$.
- 12 Consider the flow with complex potential

$$\Omega(z) = \frac{1}{2} \left[\frac{e^z + 1}{e^z - 1} \right].$$

Use a computer to draw the streamlines, and verify mathematically that the picture may be interpreted as the flow that results when the dipole with complex potential $\Omega(z) = (1/z)$ is confined to the channel $-\pi \leq \text{Im}(z) \leq \pi$.

- 13 Continuing from the previous question, what would the new complex potential be if fluid were flowing down the channel with speed v prior to the insertion of the dipole? Check your answer by using a computer to draw the streamlines.
- 14 Suppose that the doublet consisting of a source of strength 2π at $z = 1$ and a sink of equal strength at $z = -1$ is inserted into the uniform flow with real, positive velocity v . Locate the “stagnation points” (singular points of zero velocity) of the net flow, and describe (perhaps with the aid of a computer animation) how they move as v varies from 0 to 3.
- 15 If two sources are located at opposite corners of a square, and two sinks are located at the other two corners, and all four are of equal strength, then show that the circle through these four points is a streamline. Check this by getting the computer to draw the complete flow.
- 16 Show that the streamlines produced by two vortices of equal strength are Cassinian curves (figure [2.8b], p. 68) whose foci are the locations of the two vortices. [Note that your reasoning immediately generalizes: *Cassinian curves with n foci are the streamlines of n equal vortices placed at the foci.*]
- 17 Show that the streamlines [11.15b] and the equipotentials of a quadrupole are lemniscates (see [2.9], p. 69).